

Reference Maps for Virtual Element Spaces

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1 Introduction

The finite element method (FEM) is a numerical technique for approximating solutions to partial differential equations (PDEs) by discretising a domain into cells and representing the solution locally in a finite-dimensional function space. A set of functionals (called degrees of freedom) is chosen on each grid cell, which span the dual of the local function space, and are chosen so that neighbouring cells satisfy desirable regularity conditions. Implementation requires computing a so-called nodal basis for this space: a basis of the local function space where each basis function is 1 at a distinct degree of freedom and 0 at every other. This is commonly done using a reference element, a cell upon which the basis functions are constructed once and then mapped to each cell in the mesh by a coordinate transform. By calculating and storing the basis functions only once, the implementation becomes more modular, with reduced storage and runtime costs. [13, 3] develop systematic methods for performing this mapping, enabling a wider range of techniques to be used efficiently.

The virtual element method (VEM), developed in [4], extends FEM by describing local spaces through degrees of freedom and computable polynomial projections. At the same time, the associated basis exists abstractly but is not used in its closed form. By projecting the basis functions, the local spaces are computationally decoupled from the degrees of freedom, meaning that the method can use the degrees of freedom to compute the necessary projected quantities without needing an explicit nodal basis. Previously, these projections depended on the type of PDE problem considered, so VEM did not easily extend to, for example, non-linear problems. [10] introduced a more general framework to solve this shortcoming by formulating a constrained least-squares problem on each cell.

Unlike standard FEM, VEM does not, in general, admit a straightforward reference-element formulation. This project extends the reference-element mapping of [13, 3] to triangular virtual element spaces within the constrained least-squares style of [10]. We illustrate the theory for Lagrange-type and Hermite-type virtual element spaces and apply this construction to solve a model Poisson problem.

Section 2 introduces the mesh, function space, and mapping notation used throughout. Section 3 reviews the finite element theory of [13, 3]. Section 4 develops the virtual element framework and derives the main mapping results. Section 5 presents two concrete VEM spaces, modelled after the Lagrange and Hermite families of finite elements. Finally, Section 6 demonstrates [11], an open source implementation written to accompany this document.

Where definitions and examples are taken from papers, they will be provided alongside a reference, and the same holds for results, where the proofs will also be referenced unless they were omitted, in which case an alternative proof will be provided. Results and proofs stated without reference are thought to be original by the author.

2 Preliminaries and Reference Maps

2.1 Mesh notation

The elements under study are triangles in two dimensions, so that for any pair of triangles, there is an affine-linear map between them.

Definition 2.1 (Mesh [10]). A mesh $\mathcal{T}_h$ is a tessellation of a polygonal domain $\Omega \subset \mathbb{R}^2$ into triangular elements which do not overlap, where each edge is either part of the boundary,

or shared with another element in $\mathcal{T}_h$.

Definition 2.2 (Element [10]). An element in $\mathcal{T}_h$ is denoted E , and $h_E := \text{diam}(E)$. We let $h := \max_{E \in \mathcal{T}_h} \{h_E\}$.

We set the following conventions on each element:

1. Each element is a collection of vertices (v_1, v_2, v_3) which are positively oriented ($v_1 \rightarrow v_2 \rightarrow v_3$ forms an anti-clockwise loop).
2. The edge e_i connects the vertices which are not v_i .
3. The unit normal n_i to edge e_i is defined to be outward-pointing.
4. The unit tangent t_i to edge e_i is Rn_i , where

$$R = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \quad (2.1)$$

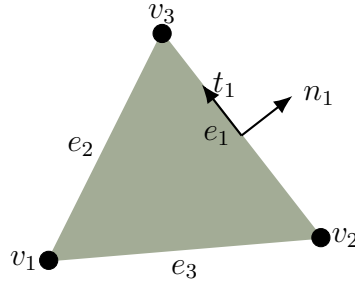


Figure 1: A triangular element with v_1 chosen arbitrarily, and the remaining vertices, edges, tangents and normals conforming to our conventions.

Definition 2.3 (Mesh interface [10]). We denote by $\mathcal{S}_h$ the set of all mesh interfaces, or edges, contained in $\mathcal{T}_h$. We define $\mathcal{S}_h(E) = \{e_i\}_{i=1}^3$.

Note that two elements sharing an edge e will have different normals and tangents defined on e . This does not affect our local study and can be fixed during global mesh assembly.

2.2 Function space notation

We now fix an element $E \in \mathcal{T}_h$, and consider spaces of functions on E .

Definition 2.4 (Local ambient space). The local ambient space W_E is a real vector space of functions on E such that for every $v_h \in W_E$, $v_h|_e$ is well defined for every $e \in \mathcal{S}_h(E)$.

Remark 2.5 (Edge restrictions). The phrase “ $v_h|_e$ is well defined” is understood in the sense appropriate to the ambient space. For example, if $W_E \subset \mathcal{C}^0(\bar{E})$, then restriction is pointwise. If $W_E \subset H^2(E)$, $v_h|_e$ is understood as the Sobolev trace on e . Equality on e means equality of the corresponding traces.

Definition 2.6 (Degrees of freedom [10]). We define the set of degrees of freedom (dofs) $\Lambda^E = (\lambda_i)_{i=1}^{N_E}$ as a set of elements of the dual space, W'_E .

On this dof set, for $e \in \mathcal{S}_h(E)$, we define the subset

$$\Lambda^{E,e} := \{\lambda_i : \lambda_i(v_h) = \lambda_i(w_h) \text{ if } v_h|_e = w_h|_e\}. \quad (2.2)$$

Remark 2.7. Fix a finite-dimensional subspace $U_E \subset W_E$. Abusing notation, we again write $(\lambda_i)_{i=1}^{N_E}$ for the restricted functionals $\lambda_i|_{U_E} \in U'_E$. All subsequent notions of dual basis, computability, and unisolvency are understood with respect to this restricted dof set.

Definition 2.8 (Unisolvent set of dofs). We say that Λ^E is a unisolvent set of dofs on U_E if whenever $u \in U_E$ satisfies $\lambda_i(u) = 0$ for all i , it holds that $u = 0$.

Definition 2.9 (Point-evaluation and moment functionals). A functional $Q \in W'_E$ is of point-evaluation type if for some $x \in \bar{E}$,

$$Q(v) = v(x). \quad (2.3)$$

Q is of moment type if there is a set ω , with $\omega = E$ or $\omega = e \in \mathcal{S}_h(E)$, and a function q on ω , such that

$$Q(v) = \frac{1}{|\omega|} \int_{\omega} vq. \quad (2.4)$$

Proposition 2.10. Suppose $N_E = \dim U_E$. Then unisolvency of Λ^E on U_E is equivalent to the set $(\lambda_i)_{i=1}^{N_E}$ forming a basis of U'_E .

Proof. Let $T : U_E \rightarrow \mathbb{R}^{N_E}$ by

$$T(u) = (\lambda_1(u), \dots, \lambda_{N_E}(u)). \quad (2.5)$$

Then Λ^E is unisolvent on U_E if and only if $\ker T = \{0\}$, which, by the rank-nullity theorem, is equivalent to T being an isomorphism.

Now consider the dual map $T' : (\mathbb{R}^{N_E})' \rightarrow U'_E$ by $T'(\mu) = \mu \circ T$. Noting the convention of Remark 2.7, we make the following claim:

Claim: T' is an isomorphism if and only if $(\lambda_i)_{i=1}^{N_E}$ is a basis of U'_E .

Proof of claim: Let $(\varepsilon_i)_{i=1}^{N_E}$ be defined by

$$\varepsilon_i(x_1, \dots, x_{N_E}) = x_i, \quad (2.6)$$

forming a basis of $(\mathbb{R}^{N_E})'$. Then for any $u \in U_E$,

$$T'(\varepsilon_i)(u) = \varepsilon_i(T(u)) = \varepsilon_i(\lambda_1(u), \dots, \lambda_{N_E}(u)) = \lambda_i(u) = (\lambda_i)(u). \quad (2.7)$$

Hence, $T'(\varepsilon_i) = \lambda_i$ for each i . ■

T is an isomorphism if and only if T' is an isomorphism, so the proof is completed. □

Definition 2.11 (Dof dual basis). Suppose Λ^E is unisolvent on U_E and $N_E = \dim U_E$. We define the Λ^E -dual basis $\Phi_E = (\phi_i)_{i=1}^{N_E} \subset U_E$ by

$$\lambda_i(\phi_j) = \delta_{ij}. \quad (2.8)$$

Similarly to (2.2), we define

$$W'_{E,e} := \{Q \in W'_E : Q(v_h) = Q(w_h) \text{ if } v_h|_e = w_h|_e\}, \quad (2.9)$$

and

$$U'_{E,e} := \{Q \in U'_E : Q(v_h) = Q(w_h) \text{ if } v_h|_e = w_h|_e\}. \quad (2.10)$$

Definition 2.12 (Computable linear functional). We say $Q \in W'_E$ is computable from Λ^E if $Q \in \text{span}(\Lambda^E)$.

Similarly, for an edge $e \in \mathcal{S}_h(E)$, $Q_e \in W'_{E,e}$ is computable from $\Lambda^{E,e}$ if $Q_e \in \text{span}(\Lambda^{E,e})$.

Corollary 2.13. *Suppose Λ^E is unisolvent on U_E and $N_E = \dim U_E$. Then $Q \in W'_E$ is computable from Λ^E if and only if $Q \in U'_E$.*

Proof. By Proposition 2.10, Λ^E forms a basis for U'_E . Thus, $Q \in W'_E$ is a linear combination of elements from Λ^E if and only if $Q \in U'_E$. $\square$

We will also require a notion of computability for linear operators, which we will apply to the VEM projection operators.

Definition 2.14 (Computable linear operator). Let $\tilde{U}_E$ be a finite-dimensional space of scalar-valued functions on E and $T : \tilde{U}_E \rightarrow U_E$. We say T is computable from Λ^E if, for some basis $(m_\alpha)_{\alpha=1}^{N_E}$ of U_E , there is $(q_\alpha)_{\alpha=1}^{N_E} \in U'_E$ computable from Λ^E such that

$$T(\tilde{v}_h) = \sum_{\alpha=1}^{N_E} q_\alpha(\tilde{v}_h) m_\alpha \quad (2.11)$$

for every $\tilde{v}_h \in \tilde{U}_E$.

Similarly, for an edge $e \in \mathcal{S}_h(E)$, let $\tilde{U}_{E,e}$ be a finite-dimensional space of scalar-valued functions on e and $T_e : \tilde{U}_{E,e} \rightarrow U_{E,e}$. We say T_e is computable from $\Lambda^{E,e}$ if, for some basis $(m_\alpha^e)_{\alpha=1}^{N_e}$ of $U_{E,e}$, there is $(q_\alpha^e)_{\alpha=1}^{N_e} \in U'_{E,e}$ computable from $\Lambda^{E,e}$ such that

$$T_e(\tilde{v}_h) = \sum_{\alpha=1}^{N_e} q_\alpha^e(\tilde{v}_h) m_\alpha^e \quad (2.12)$$

for every $\tilde{v}_h \in \tilde{U}_{E,e}$.

Definition 2.15 (Polynomials and basis). The space of polynomials over $E \in \mathcal{T}_h$ of degree at most $\ell \geq 0$ is denoted $\mathbb{P}_\ell(E)$. We require a hierarchical basis for these polynomials, which we denote by $\mathcal{M}_\ell(E)$.

It will be useful to introduce a similar notion for the element edges.

Definition 2.16 (Boundary space [18]). For $\ell \geq 0$, we define the boundary space on the triangular element E to be

$$\mathbb{B}_\ell(\partial E) := \{v \in C^0(\partial E) : v|_{e_i} \in \mathbb{P}_\ell(e_i), i \in \{1, 2, 3\}\}. \quad (2.13)$$

Example 2.17 (Scaled monomials [10]). For $k \in \mathbb{N}$, we define the scaled monomials of order k as

$$\mathcal{M}_k^*(E) = \left\{ \left(\frac{x - x_E}{h_E} \right)^\alpha : \alpha \in \mathbb{N}^2, |\alpha| = k \right\}, \quad (2.14)$$

where x_E denotes the barycentre of the element E and α a multi-index with $|\alpha| = \sum_{i=1}^2 \alpha_i$. Then,

$$\mathcal{M}_\ell(E) := \bigcup_{k \leq \ell} \mathcal{M}_k^*(E) \quad (2.15)$$

is a basis of $\mathbb{P}_\ell(E)$.

We can construct polynomial bases on edges $e \in \mathcal{S}_h(E)$ in the same way as elements to build $\mathbb{B}_\ell(e)$.

2.3 Reference maps

Numerical schemes often require evaluating nodal basis functions and their derivatives on each mesh element. Computing these basis functions on every element can be computationally expensive; therefore, it is common to implement them for a reference element and use a map to reconstruct them on a given element, referred to as the physical element.

Let us denote the reference element $\hat{E}$. Hats denote quantities on the reference element. For a physical element $E \in \mathcal{T}_h$, let $F : \hat{E} \rightarrow E$ be a diffeomorphism. In particular, for triangular elements, F is an affine-linear map of the form

$$F(\hat{x}) = J\hat{x} + b, \quad (2.16)$$

whereby our assumption that the elements are positively oriented ensures $\det J > 0$. This induces a mapping between functions and, in turn, a mapping between functionals. Let $W_{\hat{E}}$ and W_E be local vector spaces of functions on $\hat{E}$ and E respectively. Table 1 contains a summary of mapping notation from this section for reference during the later chapters.

2.3.1 Pull-back and push-forward

Definition 2.18 (Pull-back [3]). The pull-back operation mapping $W_{\hat{E}}$ to W_E is defined by

$$F^*(\hat{f}) := \hat{f} \circ F^{-1}, \quad (2.17)$$

for each $\hat{f} \in W_{\hat{E}}$.

Definition 2.19 (Push-forward [3]). The push-forward operation maps the dual space W'_E to $W'_{\hat{E}}$ by

$$F_*(\lambda) := \lambda \circ F^*, \quad (2.18)$$

for each $\lambda \in W'_E$.

We will not require these maps to be defined on the full infinite-dimensional function and dual spaces, so we will only consider their restriction to finite-dimensional subspaces $U_{\hat{E}}$ and U_E .

Proposition 2.20. *Suppose that $F^*(U_{\hat{E}}) = U_E$. Then $F^* : U_{\hat{E}} \rightarrow U_E$ is a vector-space isomorphism.*

Consequently, the induced map $F_ : U'_E \rightarrow U'_{\hat{E}}$ by $F_*(\lambda) = \lambda \circ F^*$ is also a vector space isomorphism.*

Proof. Linearity follows since

$$(\alpha v_h + \beta w_h) \circ F^{-1} = \alpha(v_h \circ F^{-1}) + \beta(w_h \circ F^{-1}). \quad (2.19)$$

Suppose $F^*(v_h) = v_h \circ F^{-1} = 0$. Then $v_h = 0 \circ F = 0$, so the pull-back is injective. Since surjectivity is assumed, F^* is a vector-space isomorphism. An identical argument for $F_* : U'_E \rightarrow U'_{\hat{E}}$ gives the result. $\square$

We refer to the inverse pull-back and push-forward as $F^{-*} : U_E \rightarrow U_{\hat{E}}$ and $F_{-*} : U'_{\hat{E}} \rightarrow U'_E$, respectively. Under these maps, the functionals from [Definition 2.9](#) push-forward to their corresponding functional in the reference dual space, making them useful degrees of freedom.

Lemma 2.21 (Point-evaluation and moment functionals under push-forward). *Suppose $\delta_{x_0} \in W'_E$ is the point-evaluation type functional which evaluates a function at a point $x_0 \in \bar{E}$, and $\hat{x}_0 := F^{-1}(x_0) \in \bar{\hat{E}}$. Then $F_*(\delta_{x_0}) = \delta_{\hat{x}_0} \in W'_{\hat{E}}$.*

Let $Q \in W'_E$ be a moment type functional given by

$$Q(v) = \frac{1}{|\omega|} \int_{\omega} v q \quad (2.20)$$

for a set ω , with $\omega = E$ or $\omega = e \in \mathcal{S}_h(E)$, and a function q on ω . Defining $\hat{\omega} = F^{-1}(\omega)$ and $\hat{q} := F^{-}(q)$, it holds that*

$$F_*(Q)(\hat{v}) = \frac{1}{|\hat{\omega}|} \int_{\hat{\omega}} \hat{v} \hat{q}. \quad (2.21)$$

We say that these functionals push forward trivially.

Proof. For the point-evaluation functional, we have

$$F_*(\delta_{x_0})(\hat{v}) = \delta_{x_0}(F^*(\hat{v})) = \delta_{x_0}(\hat{v} \circ F^{-1}) = \delta_{\hat{x}_0}(\hat{v}), \quad (2.22)$$

for any $\hat{v} \in W_{\hat{E}}$. For the moment type functional, define $\hat{q} = F^{-*}(q)$ and take $\hat{v} \in W_{\hat{E}}$. Then,

$$F_*(Q)(\hat{v}) = \frac{1}{|\omega|} \int_{\omega} F^*(\hat{v}) q = \frac{1}{|\omega|} \int_{\hat{\omega}} \hat{v} \hat{q} \frac{|\omega|}{|\hat{\omega}|} = \frac{1}{|\hat{\omega}|} \int_{\hat{\omega}} \hat{v} \hat{q}. \quad (2.23)$$

$\square$

The following corollary forms a connection between dof sets on related elements.

Corollary 2.22. *Let $\Lambda^E = (\lambda_i)_{i=1}^{N_E} \subset U'_E$. Then Λ^E is unisolvent on U_E if and only if*

$$F_*(\Lambda^E) := (F_*(\lambda_i))_{i=1}^{N_E} \quad (2.24)$$

is unisolvent on $U_{\hat{E}}$.

Proof. Assume first that Λ_E is unisolvent on U_E . Let $\hat{u} \in U_{\hat{E}}$ satisfy

$$F_*(\lambda_i)(\hat{u}) = 0 \quad (2.25)$$

for $i = 1, \dots, N_E$. By definition,

$$F_*(\lambda_i)(\hat{u}) = \lambda_i(F^*(\hat{u})), \quad (2.26)$$

so $\lambda_i(F^*(\hat{u})) = 0$ for all i . By unisolvency, $F^*(\hat{u}) = 0$, and by injectivity of F^* , $\hat{u} = 0$.

Conversely, assume $F^*(\Lambda^E)$ is unisolvent on $U_{\hat{E}}$, and let $u \in U_E$ satisfy $\lambda_i(u) = 0$ for $i = 1, \dots, N_E$. Since $F^* : U_{\hat{E}} \rightarrow U_E$ is surjective, there exists $\hat{u} \in U_{\hat{E}}$ such that $u = F^*(\hat{u})$. Then

$$F_*(\lambda_i)(\hat{u}) = \lambda_i(F^*(\hat{u})) = \lambda_i(u) = 0 \quad (2.27)$$

for all i , so by unisolvency we obtain $\hat{u} = 0$, and hence $u = 0$. $\square$

2.3.2 Piola maps

For scalar-valued functions, pull-back provides the natural transformation between reference and physical cells. However, for vector fields, the relevant degrees of freedom may depend on geometric components of the field, such as its normal or tangential component along an edge. In these cases, ordinary pull-back does not conserve desired quantities on the edge, motivating the introduction of Piola transforms. This section follows the presentation in [3].

Definition 2.23 (Contravariant Piola map [3]). Let J denote the Jacobian of F . The contravariant Piola map takes $\hat{\Psi} : \hat{E} \rightarrow \mathbb{R}^2$ to $\Psi : E \rightarrow \mathbb{R}^2$ by

$$\mathcal{F}^{\text{div}}(\hat{\Psi}) := \frac{1}{\det J} J F^*(\hat{\Psi}). \quad (2.28)$$

Remark 2.24. We would generally take the absolute value of the determinant $\det J$ in (2.28); [17] details the choice of sign to make the mapping work globally, but we have assumed the orientation of the elements is such that $\det J > 0$.

Proposition 2.25 (Contravariant conservation of normals and integral moments [3]). *For a point x on an edge $e \in \mathcal{S}_h(E)$ with outward unit normal n , and a function $\Psi = \mathcal{F}^{\text{div}}(\hat{\Psi})$ on E , we have*

$$|e| \Psi(x) \cdot n = |\hat{e}| \hat{\Psi}(\hat{x}) \cdot \hat{n}, \quad (2.29)$$

where $\hat{e}$ and $\hat{n}$ are the corresponding edge and normal on the reference element, respectively. Consequently, the integral moments of normal components against some function $\mu(x) = \hat{\mu}(F^{-1}(x))$ along edges satisfy

$$\int_e (\Psi \cdot n) \mu \, ds = \int_{\hat{e}} (\hat{\Psi} \cdot \hat{n}) \hat{\mu} \, d\hat{s}. \quad (2.30)$$

Proof. [3] We begin by noting that, if e has end points x_i and x_j , then

$$x_i - x_j = F(\hat{x}_i) - F(\hat{x}_j) = J(\hat{x}_i - \hat{x}_j), \quad (2.31)$$

so J maps $\hat{e}$ onto e . Therefore

$$t = \frac{|\hat{e}|}{|e|} J\hat{t}. \quad (2.32)$$

Using $\frac{1}{\det J} RJR^T = J^{-T}$ for 2x2 matrices, where R is as in (2.1), we have

$$|e| n = |e| R\hat{t} = |\hat{e}| RJ\hat{t} = |\hat{e}| (\det J) J^{-T} R\hat{t} = |\hat{e}| (\det J) J^{-T} R\hat{n}. \quad (2.33)$$

Applying these identities,

$$|e| \Psi(x) \cdot n = \left(\frac{1}{\det J} J\hat{\Psi}(\hat{x}) \right)^T (|\hat{e}| (\det J) J^{-T} \hat{n}) \quad (2.34)$$

$$= |\hat{e}| \hat{\Psi}(\hat{x}) \cdot \hat{n}. \quad (2.35)$$

The coordinate transform from e to $\hat{e}$ contributes stretch factor of $|e|/|\hat{e}|$ so under the above equality the integral becomes

$$\int_e (\Psi \cdot n) \mu \, ds = \int_{\hat{e}} \frac{|\hat{e}|}{|e|} \left(\hat{\Psi} \cdot \hat{n} \right) \hat{\mu} \frac{|e|}{|\hat{e}|} \, d\hat{s} \quad (2.36)$$

$$= \int_{\hat{e}} \left(\hat{\Psi} \cdot \hat{n} \right) \hat{\mu} \, d\hat{s}, \quad (2.37)$$

giving the desired integral equality. $\square$

We have a similar mapping for spaces with tangential degrees of freedom, such as those discretising $H(\text{curl})$, via the Covariant Piola map:

Definition 2.26 (Covariant Piola map [3]). The covariant Piola map takes $\hat{\Psi} : \hat{E} \rightarrow \mathbb{R}^2$ to $\Psi : E \rightarrow \mathbb{R}^2$ by

$$\mathcal{F}^{\text{curl}}(\hat{\Psi}) := J^{-T} F^*(\hat{\Psi}). \quad (2.38)$$

Example 2.27. For any differentiable $\hat{u}$ defined on $\hat{E}$, the chain rule gives

$$\nabla F^*(\hat{u}) = \nabla(\hat{u} \circ F^{-1}) = J^{-T} \hat{\nabla}(\hat{u}) \circ F^{-1} = J^{-T} F^*(\hat{\nabla} \hat{u}) = \mathcal{F}^{\text{curl}}(\hat{\nabla} \hat{u}), \quad (2.39)$$

so the covariant Piola map gives the correct transformation law for gradients under pull-back.

Proposition 2.28 (Covariant conservation of tangents [3]). For a point x on an edge $e \in \mathcal{S}_h(E)$ and a function $\Psi = \mathcal{F}^{\text{curl}}(\hat{\Psi})$ on E , it holds

$$|e| \Psi(x) \cdot t = |\hat{e}| \hat{\Psi}(\hat{x}) \cdot \hat{t}. \quad (2.40)$$

Proof. [3] Using (2.32), we have

$$\Psi(x) \cdot t = \left(J^{-T} \hat{\Psi}(\hat{x}) \right)^T \frac{|\hat{e}|}{|e|} J\hat{t} = \frac{|\hat{e}|}{|e|} \hat{\Psi}(\hat{x}) \cdot \hat{t}, \quad (2.41)$$

which proves the claim. $\square$

3 Classical Finite-Element Construction

Using the framework from [13] and the later [3], we now define finite elements and equivalence properties between elements, demonstrating the mappings from the previous section. Exposing the underlying mapping structure between finite elements will allow us to draw parallels once we define the more general virtual element methods, helping to extend the mapping methodology.

3.1 Finite elements: definition and properties

Let $\mathcal{C}^k(\bar{E})$ denote the space of functions with continuous and bounded derivatives up to and including order k over the closure of E , and $\mathcal{C}^k(\bar{E})'$ its topological dual.

Definition 3.1 (Ciarlet finite element [8, 13]). A finite element is a triple (E, P^E, Λ^E) such that

- $E \subset \mathbb{R}^d$ is a bounded domain.
- $P^E \subset \mathcal{C}^k(\bar{E})$ for some $k \geq 0$, with $\dim P^E = N_E < \infty$.
- $\Lambda^E = (\lambda_i)_{i=1}^{N_E} \subset \mathcal{C}^k(\bar{E})'$ is a collection of linearly independent functionals whose actions restricted to P^E form a basis for the dual space $(P^E)'$.

We refer to E as the element and P^E as the shape function space. By [Proposition 2.10](#), Λ^E forms a unisolvent set of degrees of freedom on P^E .

Remark 3.2 (Nodal basis). Let (E, P^E, Λ^E) be a finite element. The nodal basis Φ for P^E is the Λ^E -dual basis on P^E , defined in [Definition 2.11](#). That is, $\Phi = (\phi_i)_{i=1}^{N_E} \in (P^E)^{N_E}$ such that $\lambda_i(\phi_j) = \delta_{ij}$ for all i, j .

3.2 Notions of equivalence

We now fix a reference finite element, and relax the notation slightly, denoting it $(\hat{E}, \hat{P}, \hat{\Lambda})$. Similarly, we shall study a single, generic physical finite element, denoted (E, P, Λ) .

Let $F : \hat{E} \rightarrow E$ be an affine map between these elements.

Definition 3.3 (Affine equivalence [13]). $(\hat{E}, \hat{P}, \hat{\Lambda})$ and (E, P, Λ) are affine equivalent if

- $F(\hat{E}) = E$,
- The pull-back maps $F^*(\hat{P}) = P$,
- The push-forward maps $F_*(\hat{\Lambda}) = \Lambda$.

Definition 3.4 (Nodal interpolant, interpolation equivalent, affine-interpolation equivalent [13]). Let Φ be the nodal basis of the physical element. The nodal interpolant $\mathcal{I}_\Lambda : \mathcal{C}^k(\bar{E}) \rightarrow P$ is defined by

$$\mathcal{I}_\Lambda(f) = \sum_{i=1}^{N_E} \lambda_i(f) \phi_i. \quad (3.1)$$

Two finite elements (E, P, Λ) and $(E, P, \tilde{\Lambda})$ are said to be interpolation equivalent if $\mathcal{I}_\Lambda = \mathcal{I}_{\tilde{\Lambda}}$.

If $(E, P, \tilde{\Lambda})$ is affine equivalent to $(\hat{E}, \hat{P}, \hat{\Lambda})$, and interpolation equivalent to (E, P, Λ) , then we call $(\hat{E}, \hat{P}, \hat{\Lambda})$ and (E, P, Λ) affine-interpolation equivalent.

Proposition 3.5. $\lambda_i(\mathcal{I}_\Lambda f) = \lambda_i(f)$ for every $f \in \mathcal{C}^k(\bar{E})$.

Proof. We have

$$\lambda_i(\mathcal{I}_\Lambda f) = \lambda_i \left(\sum_{j=1}^{N_E} \lambda_j(f) \phi_j \right) = \sum_{j=1}^{N_E} \lambda_j(f) \lambda_i(\phi_j) = \lambda_i(f). \quad (3.2)$$

□

Proposition 3.6. [13] Finite elements (E, P, Λ) and $(E, P, \tilde{\Lambda})$ are interpolation equivalent if and only if the spans of Λ and $\tilde{\Lambda}$, viewed as subsets of $\mathcal{C}^k(\bar{E})'$, are equal.

Proof. The “only if” direction is proved in [6].

(Only if) We must show that $\mathcal{I}_\Lambda f = \mathcal{I}_{\tilde{\Lambda}} f$ for every $f \in \mathcal{C}^k(\bar{E})$. Assuming that the spans of Λ and $\tilde{\Lambda}$ are equal, for each λ_i we can write

$$\lambda_i = \sum_{j=1}^{N_E} c_{ij} \tilde{\lambda}_j. \quad (3.3)$$

Therefore, applying Proposition 3.5 twice,

$$\lambda_i(\mathcal{I}_{\tilde{\Lambda}} f) = \left(\sum_{j=1}^{N_E} c_{ij} \tilde{\lambda}_j \right) (\mathcal{I}_{\tilde{\Lambda}} f) = \sum_{j=1}^{N_E} c_{ij} \tilde{\lambda}_j(f) = \lambda_i(f) = \lambda_i(\mathcal{I}_\Lambda f), \quad (3.4)$$

thus $\lambda_i(\mathcal{I}_{\tilde{\Lambda}} f - \mathcal{I}_\Lambda f) = 0$. Unisolvency of Λ proves the result.

(if) We now assume that (E, P, Λ) and $(E, P, \tilde{\Lambda})$ are interpolation equivalent, so $\mathcal{I}_\Lambda f = \mathcal{I}_{\tilde{\Lambda}} f$ for every $f \in \mathcal{C}^k(\bar{E})$. Since $\tilde{\Lambda}$ forms a basis of P' , we can write

$$\lambda_i(p) = \sum_{j=1}^{N_E} c_{ij} \tilde{\lambda}_j(p) \quad (3.5)$$

for every $p \in P$. For $f \in \mathcal{C}^k(\bar{E})$, Proposition 3.5 and interpolation equivalence give

$$\lambda_i(f) = \lambda_i(I_\Lambda f) = \lambda_i(I_{\tilde{\Lambda}} f). \quad (3.6)$$

Since $I_{\tilde{\Lambda}} f \in P$,

$$\lambda_i(f) = \lambda_i(I_{\tilde{\Lambda}} f) = \sum_{j=1}^{N_E} c_{ij} \tilde{\lambda}_j(I_{\tilde{\Lambda}} f) = \sum_{j=1}^{N_E} c_{ij} \tilde{\lambda}_j(f). \quad (3.7)$$

As the above holds for every $f \in \mathcal{C}^k(\bar{E})$, we have that each λ_i is a linear combination of members of $\tilde{\Lambda}$ in P' , proving the claim. □

3.3 Mapping theory

For now, we assume that F correctly converts the reference function space into the physical function space, i.e. $F^*(\hat{P}) = P$, allowing us to discuss the relationship between the reference nodes $\hat{\Lambda}$ and the pushed-forward physical nodes $F_*(\Lambda)$. Many finite element methods use polynomial basis functions, so we first show that polynomial spaces satisfy this property.

Lemma 3.7 (Affine pull-back of polynomial spaces). *Let $F : \hat{E} \rightarrow E$ be an invertible affine mapping. Then for any $k \geq 0$,*

$$F^*(\mathbb{P}_k(\hat{E})) = \mathbb{P}_k(E). \quad (3.8)$$

Proof. Let $\hat{p} \in \mathbb{P}_k(\hat{E})$. Since F^{-1} is affine, each component of $F^{-1}(x)$ is a linear function of x . Therefore $F^*(\hat{p})$ has degree at most k . Hence $F^*(\mathbb{P}_k(\hat{E})) \subset \mathbb{P}_k(E)$. Conversely, for any $p \in \mathbb{P}_k(E)$ define $\hat{p} := F^{-*}(p)$. Since F is affine, $\hat{p} \in \mathbb{P}_k(\hat{E})$, and clearly $F^*(\hat{p}) = p$, so $\mathbb{P}_k(E) \subset F^*(\mathbb{P}_k(\hat{E}))$. $\square$

Since we have mapped the local shape function space from the reference element, it remains to understand how the degrees of freedom transform under push-forward.

Definition 3.8. [13] Let $\Lambda = (\lambda_i)_{i=1}^{N_E}$ be a collection of functionals, and $\Psi = (\psi_i)_{i=1}^\ell$ be a collection of functions. We define the (outer) product $\Lambda(\Psi)$ to be the $k \times \ell$ matrix

$$(\Lambda(\Psi))_{ij} = \lambda_i(\psi_j). \quad (3.9)$$

Lemma 3.9. [13] Let $\Psi \in P^{N_E}$ and $M \in \mathbb{R}^{N_E \times N_E}$. Then

$$\Lambda(M\Psi) = \Lambda(\Psi)M^T. \quad (3.10)$$

Proof. [13] By (3.9) and linearity of λ_i ,

$$(\Lambda(M\Psi))_{ij} = \lambda_i \left(\sum_{k=1}^{N_E} M_{jk} \psi_k \right) = \sum_{k=1}^{N_E} \lambda_i(\psi_k) M_{jk} = \sum_{k=1}^{N_E} (\Lambda(\Psi))_{ik} M_{jk}, \quad (3.11)$$

and the right-hand side is $\Lambda(\Psi)M^T$. $\square$

Proposition 3.10. [13] Let $\Phi = \{\phi_i\}_{i=1}^{N_E}$ and $\hat{\Phi} = \{\hat{\phi}_i\}_{i=1}^{N_{\hat{E}}}$ be the nodal bases for finite elements (E, P, Λ) and $(\hat{E}, \hat{P}, \hat{\Lambda})$, respectively. Then under the assumption that $F^*(\hat{P}) = P$ and $F(\hat{E}) = E$, there exists an invertible $N_E \times N_E$ matrix M such that

$$\Phi = MF^*(\hat{\Phi}). \quad (3.12)$$

Proof. [13] The pull-back is invertible, so it maps the linearly independent set $\hat{\Phi}$ to a linearly independent set. Since this set is also a basis for P , a change-of-basis matrix exists (and is unique). $\square$

Define the intermediate $N_E \times N_E$ matrix $B = F_*(\Lambda)(\hat{\Phi})$. By Definition 3.8 of the outer product, the entries are

$$B_{ij} = F_*(\lambda_i)(\hat{\phi}_j) \quad (3.13)$$

$$= \lambda_i \left(F^*(\hat{\phi}_j) \right). \quad (3.14)$$

Since the push-forward operation is an isomorphism, the nodal sets $\hat{\Lambda}$ and $F_*(\Lambda)$ are both bases for $\hat{P}'$, so there exists an invertible $N_E \times N_E$ matrix V such that

$$\hat{\Lambda} = VF_*(\Lambda). \quad (3.15)$$

In practice, it is often easier to express the pushed-forward nodes as a linear combination of the reference nodes, producing the inverse matrix V^{-1} . Having calculated V from the reference nodes, the following result yields the nodal basis map, M .

Theorem 3.11. [13] Let (E, P, Λ) and $(\hat{E}, \hat{P}, \hat{\Lambda})$ be finite elements with $F(\hat{E}) = E$ and $F^*(\hat{P}) = P$. Then

$$M = V^T. \quad (3.16)$$

Proof. [13] By the Kronecker property of nodal bases,

$$I = \Lambda(\Phi) = \Lambda(MF^*(\hat{\Phi})) = \Lambda(F^*(\hat{\Phi}))M^T = BM^T, \quad (3.17)$$

so $M = B^{-T}$. Similarly,

$$I = (VF_*(\Lambda))(\hat{\Phi}) = VF_*(\Lambda)(\hat{\Phi}) = VB, \quad (3.18)$$

so that $V = B^{-1}$, and the result follows. $\square$

Since the affine-equivalence of two finite elements requires $F_*(\Lambda) = \hat{\Lambda}$, the following corollary is immediate.

Corollary 3.12. [13] If (E, P, Λ) and $(\hat{E}, \hat{P}, \hat{\Lambda})$ are affine-equivalent finite elements, then the transformation M is the identity matrix.

Proof.

$$\Lambda(F^*(\hat{\Phi})) = F_*(\Lambda)(\hat{\Phi}) = \hat{\Lambda}(\hat{\Phi}) = I. \quad (3.19)$$

$\square$

Example 3.13 (Affine-equivalence: the Lagrange finite element, [13, $k = 1$ case]). The Lagrange finite element of order k on reference element $\hat{E}$ is defined by the triple $(\hat{E}, \mathbb{P}_k(\hat{E}), \hat{\Lambda})$, where $\hat{\lambda}_i = \delta_{\hat{x}_i}$ for a collection of $N_{\hat{E}} := \frac{1}{2}(k+1)(k+2)$ points $\{\hat{x}_i\}$.

For a physical element E , we define $\lambda_i = \delta_{x_i}$, where $x_i = F(\hat{x}_i)$. Then $(\hat{E}, \mathbb{P}_k(\hat{E}), \hat{\Lambda})$ and $(E, \mathbb{P}_k(E), \Lambda)$ are affine-equivalent [6], so by Corollary 3.12, we have that the matrix M is the identity. In fact, this is easy to see since for any $\hat{p} \in \mathbb{P}_k(\hat{E})$, Lemma 2.21 gives

$$F_*(\lambda_i)(\hat{p}) = \hat{\lambda}_i(\hat{p}). \quad (3.20)$$

That is,

$$F_*(\Lambda) = \hat{\Lambda}, \quad (3.21)$$

so $M = V = I$. Consequently, the implementation of these finite elements can be simplified by evaluating the nodal basis on the reference element and mapping by pull-back, exploiting the relationship

$$\Phi = F^*(\hat{\Phi}). \quad (3.22)$$

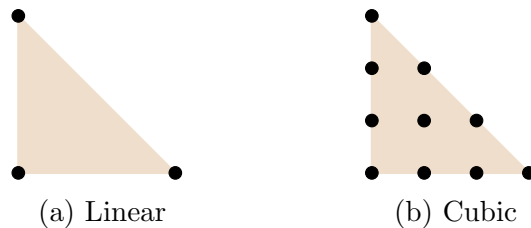


Figure 2: Reference Lagrange-type finite elements.

We now include an example of non-trivial behaviour under pull-back, induced by the inclusion of derivative nodes.

Example 3.14 (Affine-interpolation equivalence: the cubic Hermite finite element [13]). Let $\hat{E}$ be the reference element with vertices $\{\hat{v}_i\}_{i=1}^3$ and barycentre $\hat{v}_4$. Then the cubic Hermite finite element is defined by the triple $(\hat{E}, \mathbb{P}_3(\hat{E}), \hat{\Lambda})$, where

$$\hat{\Lambda} = [\delta_{\hat{v}_1} \quad \nabla_{\hat{v}_1} \quad \delta_{\hat{v}_2} \quad \nabla_{\hat{v}_2} \quad \delta_{\hat{v}_3} \quad \nabla_{\hat{v}_3} \quad \delta_{\hat{v}_4}]^T. \quad (3.23)$$

That is, the degrees of freedom consist of:

1. function values at the vertices,
2. gradients at the vertices (accounting for two dofs each, one for each component),
3. a single function value at the barycentre.

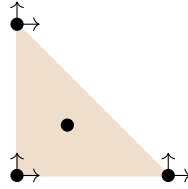


Figure 3: The reference cubic Hermite triangle.

By Lemma 2.21, the point evaluation dofs map trivially by pull-back. For any $\hat{p} \in \mathbb{P}_3(\hat{E})$, Example 2.27 showed $\nabla F^*(\hat{p}) = \mathcal{F}^{\text{curl}}(\hat{\nabla} \hat{p})$, giving an extra factor of J^{-T} . Consequently, for a physical element $(E, \mathbb{P}_3(E), \Lambda)$, $F_*(\pi \Lambda) \neq \hat{\Lambda}$, so the reference and physical elements are not affine equivalent. However, the spans of $F_*(\Lambda)$ and $\hat{\Lambda}$ are equal, so Proposition 3.6 gives affine-interpolation equivalence in the sense of Definition 3.4. Consequently, Theorem 3.11 gives $\hat{\Lambda} = V F_*(\Lambda)$ for $V = M^T \in \mathbb{R}^{10 \times 10}$. Computing V^{-1} from Lemma 2.21 and (2.39) gives the block matrix

$$F_*(\Lambda^E) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & J^{-T} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & J^{-T} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & J^{-T} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \hat{\Lambda}^E \quad (3.24)$$

$$= V^{-1} \hat{\Lambda}^E, \quad (3.25)$$

where each J^{-T} is a 2×2 matrix and each 0 is taken to be the appropriate size. Inverting, we obtain that V and, consequently, M are block-diagonal. Thus, M can be efficiently calculated and stored for each element from F , allowing us to map functions between our reference element and any element in the domain.

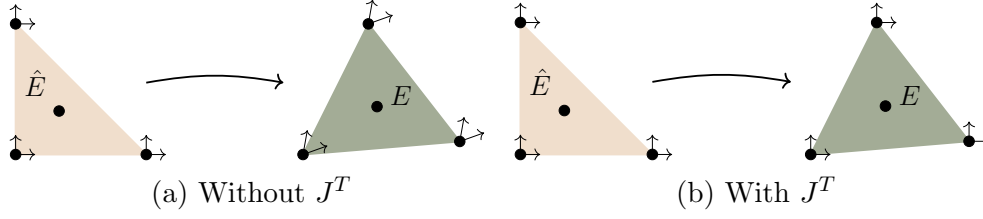


Figure 4: The J^T factor in V orients and scales each gradient node to the correct coordinate system under pull-back.

Remark 3.15. M is not always block diagonal; [13] gives the examples of the Morley [15] and Argyris [1] elements, which are not interpolation equivalent, nor do they provide a block-diagonal matrix M .

4 Virtual Elements

VEM may be viewed as a generalisation of finite element methods in which the local function space is not defined by explicit nodal basis functions, but instead by degrees of freedom and defining properties. Rather than working directly with basis functions, they are treated as virtual, meaning they exist abstractly as a dual basis but are not used in closed form. Instead, one works with polynomial projections of these virtual functions, computable from the degrees of freedom. Virtual element methods allow for the construction of high-order methods on general meshes, without the need to derive explicit functions on each element (e.g. [5, 12]). Since we would like to study mapped spaces, we will continue to restrict our attention to triangles, so the VEM framework in this section is a simplification of [10] to triangular meshes.

4.1 Virtual elements: definition and properties

The constructions in this section are local, and over an element $E \in \mathcal{T}_h$, we work with a finite-dimensional virtual space $\tilde{V}^E$. As for finite element methods, we define a set of degrees of freedom Λ^E for our element.

Assumption 4.1. [10] We assume the following

- (A1) The dof set is unisolvent on $\tilde{V}^E$, so $\Lambda^E = (\lambda_i)_{i=1}^{N_E}$, where $\dim \tilde{V}^E := N_E$.
- (A2) $\mathbb{P}_\ell(E) \subset \tilde{V}^E$, for some $\ell > 0$.

Remark 4.2. Since Λ^E is unisolvent on $\tilde{V}^E$, as in Definition 2.11, there exists a Λ^E -dual basis of $\tilde{V}^E$, but it is not generally available in closed form. We call it the virtual nodal basis and, as for finite elements, denote it Φ .

Since functions in $\tilde{V}^E$ are not computable as a linear combination of explicit nodal basis functions, we seek computable linear operators projecting from the virtual space into the polynomial space $\mathbb{P}_\ell(E)$. We denote by $\Pi_0^E[v_h]$ the value projection, which approximates v_h , and by $\Pi_1^E[v_h]$ the gradient projection, which approximates ∇v_h . Higher-order projections are possible [10], and follow hierarchically from the gradient projection via an integration-by-parts variational form.

Definition 4.3 (Virtual element tuple [10]). Define

$$\mathcal{V}^E = (E, \mathcal{B}_0^E, \Lambda^E, \mathcal{C}_0^E, \mathcal{B}_1^E), \quad (4.1)$$

consisting of:

- An element $E \in \mathcal{T}_h$.
- A basis set $\mathcal{B}_0^E$ used for value projection, with $|\mathcal{B}_0^E| \leq N_E$ and $\text{span}\mathcal{B}_0^E \subset \tilde{V}^E$.
- A set of dofs Λ^E subject to the conditions in [Assumption 4.1](#).
- A set of linear functionals $\mathcal{C}_0^E \subset \text{span}\Lambda^E$, i.e. computable from Λ^E . These will be called the constraints in what follows. We assume that the constraints are of point-evaluation type or moment type, as will be the case in the examples to come.
- A basis set $\mathcal{B}_1^E$ used for the gradient projection, with $\text{span}\mathcal{B}_1^E \subset \nabla\tilde{V}^E$.

For an edge $e \in \mathcal{S}_h$, we can define a VEM tuple $\mathcal{V}^e = (e, \mathcal{B}_0^e, \Lambda^e, \mathcal{C}_0^e)$ very similarly, except we omit a basis for gradient projection; we will only compute gradient projections in the interior of the element.

Definition 4.4 (Compatible [10]). For the set of VEM tuples $(\mathcal{V}^E)_{E \in \mathcal{T}_h}$ and $(\mathcal{V}^e)_{e \in \mathcal{S}_h}$, we say that the VEM tuples are compatible if $\Lambda^e \subset \Lambda^E$ whenever $e \in \mathcal{S}_h(E)$.

Remark 4.5. We now fix $E \in \mathcal{T}_h$ and Λ^E , and take $\Lambda^e := \Lambda^{E,e}$ for $e \in \mathcal{S}_h(E)$, ensuring compatibility. We also define the constraints $\mathcal{C}_0^e$ and basis sets $\mathcal{B}_i^e$ by restriction of $\mathcal{C}_0^E$ and $\mathcal{B}_i^E$ to e .

4.1.1 Value projection

In the consideration of possible value projections, there are three conditions under which we consider a projection to be admissible.

Definition 4.6 (Admissible element and edge value projections). We call a linear operator Π_0^E an admissible element value projection if

1. it is computable from Λ^E in the sense of [Definition 2.14](#),
2. $\Pi_0^E p = p$ for every $p \in \text{span}\mathcal{B}_0^E$,
3. $\mathcal{C}(\Pi_0^E[v_h]) = \mathcal{C}(v_h)$ for every $v_h \in \tilde{V}^E$, $\mathcal{C} \in \mathcal{C}_0^E$.

For an edge $e \in \mathcal{S}_h(E)$, we call a linear operator Π_0^e an admissible edge value projection if

1. it is computable from Λ^e in the sense of [Definition 2.14](#) for edge-valued operators,
2. $\Pi_0^e p = p$ for every $p \in \text{span}\mathcal{B}_0^e$,
3. $\mathcal{C}(\Pi_0^e[v_h]) = \mathcal{C}(v_h)$ for every $v_h \in \tilde{V}^E$, $\mathcal{C} \in \mathcal{C}_0^e$.

Example 4.7 (a_E -based value projection). *Suppose the bilinear form of the underlying weak problem has the form*

$$a(v, w) = \sum_{E \in \mathcal{T}_h} a_E(v, w), \quad (4.2)$$

where a_E represents the local bilinear form on the element E . In many virtual element constructions, one would like the polynomial projection Π_0^E to preserve the local weak structure. Accordingly, for some $v_h \in \tilde{V}^E$, a common choice in the literature ([4, 7, 12, 14, 9, 18], to name a few) is to define $\Pi_0^E[v_h] = p \in \text{span}\mathcal{B}_0^E$ by

$$a_E(v_h, q) = a_E(p, q) \quad \forall q \in \text{span}\mathcal{B}_0^E, \quad (4.3)$$

together with suitable constraints fixing the kernel (for example, matching the average over the vertices).

When a_E is a symmetric bilinear form, this condition is equivalent to choosing p to be the minimiser of the energy

$$p \mapsto a_E(v_h - p, v_h - p), \quad (4.4)$$

over the same constrained set, so in a sense, the classical projection is the polynomial projection that minimises the residual energy of the local PDE form.

This interpretation is restricted to the symmetric bilinear setting. If a_E is non-symmetric, the orthogonality condition no longer corresponds to minimisation of an energy; if a_E is non-linear, the induced projection need not be linear, so it may fail to satisfy Definition 4.6.

In standard VEM discretisations, one also introduces a stabilisation $S^E : \tilde{V}^E \times \tilde{V}^E \rightarrow \mathbb{R}$, assumed symmetric positive definite and computable from Λ^E . In addition to its role in the discrete method, S_E also provides a PDE-independent metric for defining the value projection.

Example 4.8 (Stabilisation energy minimising value projection). *Since the stabilisation $S^E(\cdot, \cdot)$ is a symmetric positive definite bilinear form, the stabilisation energy*

$$p \mapsto S^E(v_h - p, v_h - p) \quad (4.5)$$

provides an alternative residual metric. Over a feasible set defined by the constraints $\mathcal{C}_0$, we could choose the value projection to be the minimiser of this energy. The stabilisation is not unique, but a natural default is the so-called dofi-dofi stabilisation [14], given by

$$S^E(v_h, w_h) = \sum_{i=1}^{N_E} \lambda_i(v_h) \lambda_i(w_h). \quad (4.6)$$

Its value on a pair (v_h, w_h) can be evaluated solely from the degrees of freedom, yielding the Euclidean inner product in dof coordinates. In particular, the induced residual energy

$$S^E(v_h - p, v_h - p) = \sum_{i=1}^{N_E} (\lambda_i(v_h - p))^2 \quad (4.7)$$

is independent of the PDE bilinear form. For more complicated problems, such as those with varying coefficients, this leads to a simpler construction and implementation than the a_E -based projection of Example 4.7.

Definition 4.9 (Feasible set). For a fixed $v_h \in \tilde{V}^E$, we define the feasible set to be

$$\mathcal{A}_{v_h}^E := \{p \in \text{span}\mathcal{B}_0^E : \mathcal{C}(v_h) = \mathcal{C}(p), \mathcal{C} \in \mathcal{C}_0^E\}. \quad (4.8)$$

Definition 4.10 (CLS value projection [10]). For a discrete function $v_h \in \tilde{V}^E$ and an element $E \in \mathcal{T}_h$, define the value projection $\Pi_0^E[v_h]$ onto $\text{span}\mathcal{B}_0^E$ as the solution to the constrained least squares problem

$$\min \sum_{i=1}^{N_E} (\lambda_i(\Pi_0^E[v_h] - v_h))^2, \quad (4.9a)$$

subject to

$$\mathcal{C}(\Pi_0^E[v_h] - v_h) = 0 \quad \forall \mathcal{C} \in \mathcal{C}_0^E. \quad (4.9b)$$

Identically, for each edge $e \in \mathcal{S}_h(E)$, define the edge value projection $\Pi_0^e[v_h]$ onto $\text{span}(\mathcal{B}_0^e)$ as the solution to the constrained least squares problem

$$\min \sum_{i=1}^{N_e} (\lambda_i(\Pi_0^e[v_h] - v_h))^2, \quad (4.10a)$$

subject to

$$\mathcal{C}(\Pi_0^e[v_h] - v_h) = 0 \quad \forall \mathcal{C} \in \mathcal{C}_0^e. \quad (4.10b)$$

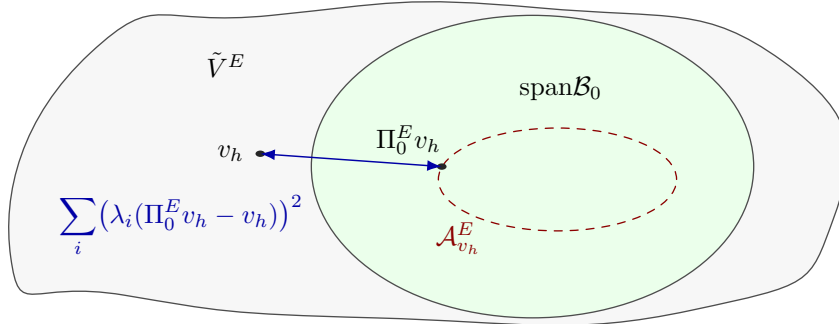


Figure 5: Schematic diagram to illustrate the role of the least-squares and constraints in the value projection of v_h into $\text{span}\mathcal{B}_0^E$.

As suggested by Figure 5, the CLS value projection of $v_h \in \tilde{V}^E$ is defined to be the closest polynomial in the feasible set, with respect to the dofi-dofi energy norm.

Remark 4.11 (L^2 -projection constraints). For optimal order convergence analysis, we want the value projections to be as close as possible to the L^2 -projection from $\tilde{V}^E$ into $\mathbb{P}_\ell$. As a result, one would typically like for the constraints to contain the set

$$\left\{ \mu \in (\tilde{V}^E)' : \mu(v) = \frac{1}{|E|} \int_E v m \, dx, m \in \text{span}\mathcal{B}_0^E \right\}, \quad (4.11)$$

or the edge equivalent, which forces the value projection to match the L^2 -projection, at least on $\text{span}\mathcal{B}_0^E \subset \mathbb{P}_\ell(E)$. We will see that higher order moment constraints are not always computable from the dofs, so often the constraints only enforce the L^2 -projection up to polynomial order $k < \ell$. This can be improved to order ℓ by the study of extended virtual element spaces, which is discussed in [10, 9].

To prove the admissibility of the value projection, we first prove that it is well-posed when the feasible set is non-empty.

Lemma 4.12 (Well-posedness of the CLS value projection). *Suppose that for each $v_h \in \tilde{V}^E$, the feasible set $\mathcal{A}_{v_h}^E$ is non-empty (where, if there are no constraints, we take $\mathcal{A}_{v_h}^E = \text{span}\mathcal{B}_0^E$). Then the constrained least-squares problem (4.9) admits a unique solution $\Pi_0^E[v_h] \in \text{span}\mathcal{B}_0^E$. Moreover, $v_h \mapsto \Pi_0^E[v_h]$ is a linear operator from $\tilde{V}^E$ to $\text{span}\mathcal{B}_0^E$ which is computable from Λ^E .*

Proof. Fix $v_h \in \tilde{V}^E$, and let $U^E := \text{span}\mathcal{B}_0^E$. For $p \in U^E$, define the objective functional

$$T_{v_h}(p) := \sum_{i=1}^{N_E} (\lambda_i(p - v_h))^2. \quad (4.12)$$

The map T_{v_h} is a continuous, quadratic functional. By definition, U^E is finite-dimensional, so $\mathcal{A}_{v_h}^E$ is a closed affine subspace of U^E , and is therefore convex. Therefore, T_{v_h} attains a minimiser on $\mathcal{A}_{v_h}^E$.

We show that this minimiser is unique by proving T_{v_h} is strictly convex on $\mathcal{A}_{v_h}^E$. To show this, let $p_1, p_2 \in \mathcal{A}_{v_h}^E$ be distinct and for $t \in [0, 1]$, consider the restriction of T_{v_h} to the line $p(t) := (1 - t)p_1 + tp_2$. Each λ_i is linear, so

$$\lambda_i(p(t) - v_h) = (1 - t)\lambda_i(p_1) + t\lambda_i(p_2) - \lambda_i(v_h) \quad (4.13)$$

$$= (1 - t)\lambda_i(p_1 - v_h) + t\lambda_i(p_2 - v_h), \quad (4.14)$$

therefore, $T_{v_h}(p(t))$ is a quadratic polynomial in t , with second derivative

$$\frac{d^2}{dt^2} T_{v_h}(p(t)) = 2 \sum_{i=1}^{N_E} (\lambda_i(p_2 - p_1))^2. \quad (4.15)$$

By [Assumption 4.1](#), unisolvency of Λ^E ensures that this second derivative is positive whenever $p_1 \neq p_2$, so T_{v_h} is strictly convex on $\mathcal{A}_{v_h}^E$, and the minimiser is unique.

For two discrete functions $v_h, w_h \in \tilde{V}^E$, let $p_{v_h} := \Pi_0^E[v_h]$ and $p_{w_h} := \Pi_0^E[w_h]$ be unique minimisers over the respective feasible sets. For $\alpha, \beta \in \mathbb{R}$, let $p := \Pi_0^E[\alpha v_h + \beta w_h]$ be the minimiser of $T_{\alpha v_h + \beta w_h}$ over $\mathcal{A}_{\alpha v_h + \beta w_h}^E$. Observe that since $\mathcal{C}(p - \alpha v_h - \beta w_h) = 0$, then by linearity of the constraints,

$$0 = \mathcal{C}(p - \alpha v_h - \beta w_h) \quad (4.16)$$

$$= \mathcal{C}(p) - \alpha \mathcal{C}(v_h) - \beta \mathcal{C}(w_h) \quad (4.17)$$

$$= \mathcal{C}(p) - \alpha \mathcal{C}(p_{v_h}) - \beta \mathcal{C}(p_{w_h}) \quad (4.18)$$

$$= \mathcal{C}(p - \alpha p_{v_h} - \beta p_{w_h}), \quad (4.19)$$

so $\alpha p_{v_h} + \beta p_{w_h} \in \mathcal{A}_{\alpha v_h + \beta w_h}^E$.

Define the constraint kernel

$$K^E := \{q \in U^E : \mathcal{C}(q) = 0, \mathcal{C} \in \mathcal{C}_0^E\}. \quad (4.20)$$

Then we may write $p = \alpha p_{v_h} + \beta p_{w_h} + k$, where $k \in K^E$. Applying the minimising property of p_{v_h} , since $p_{v_h} + tk \in \mathcal{A}_{v_h}^E$ for all $t \in \mathbb{R}$, the function $t \mapsto T_{v_h}(p_{v_h} + tk)$ has a minimum at $t = 0$. Differentiating gives

$$0 = \left. \frac{d}{dt} T_{v_h}(p_{v_h} + tk) \right|_{t=0} = 2 \sum_{i=1}^{N_E} \lambda_i(p_{v_h} - v_h) \lambda_i(k), \quad (4.21)$$

and by the same argument for p_{w_h} ,

$$0 = \sum_{i=1}^{N_E} \lambda_i(p_{w_h} - w_h) \lambda_i(k). \quad (4.22)$$

Since

$$p - (\alpha v_h + \beta w_h) = k + \alpha(p_{v_h} - v_h) + \beta(p_{w_h} - w_h), \quad (4.23)$$

expanding $T_{\alpha v_h + \beta w_h}(p)$ gives

$$T_{\alpha v_h + \beta w_h}(p) = \sum_{i=1}^{N_E} (\lambda_i(k) + \alpha \lambda_i(p_{v_h} - v_h) + \beta \lambda_i(p_{w_h} - w_h))^2 \quad (4.24)$$

$$\begin{aligned} &= \sum_{i=1}^{N_E} \lambda_i(k)^2 + \sum_{i=1}^{N_E} (\alpha \lambda_i(p_{v_h} - v_h) + \beta \lambda_i(p_{w_h} - w_h))^2 \\ &\quad + 2\alpha \sum_{i=1}^{N_E} \lambda_i(p_{v_h} - v_h) \lambda_i(k) + 2\beta \sum_{i=1}^{N_E} \lambda_i(p_{w_h} - w_h) \lambda_i(k). \end{aligned} \quad (4.25)$$

By (4.21) and (4.22) the cross terms vanish, and the second term is exactly $T_{\alpha v_h + \beta w_h}(\alpha p_{v_h} + \beta p_{w_h})$. In particular,

$$T_{\alpha v_h + \beta w_h}(p) \geq T_{\alpha v_h + \beta w_h}(\alpha p_{v_h} + \beta p_{w_h}). \quad (4.26)$$

Therefore $\alpha p_{v_h} + \beta p_{w_h}$ minimises $T_{\alpha v_h + \beta w_h}$, so uniqueness implies

$$\alpha \Pi_0^E[v_h] + \beta \Pi_0^E[w_h] = \alpha p_{v_h} + \beta p_{w_h} = p = \Pi_0^E[\alpha v_h + \beta w_h]. \quad (4.27)$$

Since the constraints are, by definition, computable from Λ^E , and the least squares problem is a function of only dof values, the projection is a linear operator computable from Λ^E in the sense of Definition 2.14. $\square$

Exactness on $\text{span} \mathcal{B}_0^E$ follows from Definition 4.10.

Lemma 4.13 (Exactness on $\text{span} \mathcal{B}_0^E$). *If $v_h \in \text{span} \mathcal{B}_0^E$, then $\Pi_0^E v_h = v_h$.*

Proof. For $v_h \in \text{span} \mathcal{B}_0^E$, the function $p = v_h$ satisfies the degrees of freedom and the constraints trivially, so it is the unique solution to the constrained least-squares problem. $\square$

Theorem 4.14 (Admissibility of the CLS value projection). *Suppose that for each $v_h \in \tilde{V}^E$, the feasible set $\mathcal{A}_{v_h}^E$ is non-empty. Then the CLS value projection (4.9) is an admissible value projection.*

Proof. We prove the three conditions from Definition 4.6.

1. By unisolvency of Λ^E on $\tilde{V}^E$, for any $v_h \in \tilde{V}^E$ we may write

$$v_h = \sum_{i=1}^{N_E} \lambda_i(v_h) \phi_i. \quad (4.28)$$

Given a non-empty feasible set $\mathcal{A}_{v_h}^E$, by [Lemma 4.12](#), Π_0^E is linear. Therefore,

$$\Pi_0^E[v_h] = \sum_{i=1}^{N_E} \lambda_i(v_h) \Pi_0^E[\phi_i]. \quad (4.29)$$

Writing each $\Pi_0^E[\phi_i] = \sum_{\alpha=1}^{N_E} c_{i,\alpha} m_\alpha$ in the basis $\mathcal{B}_0^E$, we have

$$\Pi_0^E[v_h] = \sum_{\alpha=1}^{N_E} \left(\sum_{i=1}^{N_E} c_{i,\alpha} \lambda_i(v_h) \right) m_\alpha, \quad (4.30)$$

so $\Pi_0^E[v_h]$ is computable from Λ^E in the sense of [Definition 2.14](#).

2. [Lemma 4.13](#) proves this condition exactly.
3. By definition, the value projection satisfies $\mathcal{C}(\Pi_0^E v_h) = \mathcal{C}(v_h)$ for every $v_h \in \tilde{V}^E$ and $\mathcal{C} \in \mathcal{C}_0^E$.

□

Lemma 4.15 (Admissibility of the CLS edge value projection). *Suppose that for each $v_h \in \tilde{V}^E$, the feasible set*

$$\mathcal{A}_{v_h}^e := \{p \in \text{span} \mathcal{B}_0^e : \mathcal{C}(p - v_h) = 0, \mathcal{C} \in \mathcal{C}_0^e\} \quad (4.31)$$

is non-empty. Then the CLS edge value projection defined in [Definition 4.10](#) is linear, exact on $\text{span} \mathcal{B}_0^e$, and is an admissible edge value projection.

Proof. The proof is identical to the proofs of [Lemma 4.12](#), [Lemma 4.13](#), and [Theorem 4.14](#), with the element quantities $(\mathcal{B}_0^E, \Lambda^E, \mathcal{C}_0^E)$ replaced by the corresponding edge quantities $(\mathcal{B}_0^e, \Lambda^e, \mathcal{C}_0^e)$. □

4.1.2 Constructing the value projection

The following construction is detailed in [\[10\]](#).

For an element $E \in \mathcal{T}_h$ we would like to compute the value projections of the nodal basis

$$\Pi_0^E(\Phi^E) := (\Pi_0^E[\phi_k^E])_k \quad (4.32)$$

in the span of $\mathcal{B}_0^E$. Define $\mathcal{N}_0^E := \dim \mathcal{B}_0^E \leq N^E$ as the size of the basis set and write $\mathcal{B}_0^E = (m_\alpha^E)_\alpha \subset \mathbb{P}_\ell(E)$, picking an arbitrary ordering for the basis set. Since m_α^E spans the range space of the value projection, we can construct the matrix $\mathbf{\Pi}_0^E$ to provide the coefficients of the projection in the span of the basis set

$$\Pi_0^E[\phi_k^E] = \sum_{\alpha=1}^{\mathcal{N}_0^E} (\mathbf{\Pi}_0^E)_{\alpha,k} m_\alpha^E. \quad (4.33)$$

That is, the k -th column of $\mathbf{\Pi}_0^E$ contains the coefficients of the polynomial $\Pi_0^E[\phi_k^E]$ in the basis $(m_\alpha^E)_\alpha$.

Noting linearity of the λ_i^E , we can rewrite the constrained least squares problem from [Definition 4.10](#) in Π_0^E as follows

$$\sum_{i=1}^{N^E} (\lambda_i^E (\Pi_0^E [\phi_k^E]) - \lambda_i^E (\phi_k^E))^2 = \sum_{i=1}^{N^E} \left(\sum_{\alpha=1}^{\mathcal{N}_0^E} (\Pi_0^E)_{\alpha,k} \lambda_i^E (m_\alpha^E) - \lambda_i^E (\phi_k^E) \right)^2 \quad (4.34)$$

$$= \left\| \tilde{A}^E (\Pi_0^E)_k - e_k \right\|^2 \quad (4.35)$$

where $\tilde{A}_{i,\alpha}^E = \lambda_i^E (m_\alpha^E)$ is the dof evaluation matrix, $(\Pi_0^E)_k$ is the k -th column of the matrix Π_0^E , and e_k is the Kronecker delta vector. We can set up the constraints system as follows

$$0 = \mathcal{C} (\Pi_0^E [\phi_k^E] - \phi_k^E) \quad (4.36)$$

$$= \mathcal{C} \left(\sum_{\alpha=1}^{\mathcal{N}_0^E} (\Pi_0^E)_{\alpha,k} m_\alpha^E - \phi_k^E \right) \quad (4.37)$$

$$= \sum_{\alpha=1}^{\mathcal{N}_0^E} (\Pi_0^E)_{\alpha,k} \mathcal{C} (m_\alpha^E) - \mathcal{C} (\phi_k^E) \quad (4.38)$$

where we have assumed linearity of $\mathcal{C} \in \mathcal{C}_0^E$. Ordering the constraints into a vector of length $\mathcal{N}_*^E \leq \mathcal{N}_0^E$, also denoted $\mathcal{C}_0^E = (\mathcal{C}_j)_{j=1}^{\mathcal{N}_*^E}$, and rewriting as a matrix system gives

$$\tilde{C}^E (\Pi_0^E)_k - \mathcal{C}_0 (\phi_k^E) = 0, \quad (4.39)$$

where $\tilde{C}_{j\alpha}^E = \mathcal{C}_j (m_\alpha)$. Then for each $k = 1, \dots, N^E$, we may introduce Lagrange multipliers ν for the linear constraints, and define the Lagrangian [\[16\]](#)

$$\mathcal{L}(x, \nu) = \frac{1}{2} \left\| \tilde{A}^E x - e_k \right\|^2 + \nu^T \left(\tilde{C}^E x - \mathcal{C}_0 (\phi_k^E) \right). \quad (4.40)$$

This Lagrangian gives rise to the following KKT system, which characterises the constrained least squares problem:

$$\begin{bmatrix} (\tilde{A}^E)^T \tilde{A}^E & (\tilde{C}^E)^T \\ \tilde{C}^E & 0 \end{bmatrix} \begin{bmatrix} (\Pi_0^E)_k \\ \nu \end{bmatrix} = \begin{bmatrix} (\tilde{A}^E)^T e_k \\ \mathcal{C}_0 (\phi_k^E) \end{bmatrix}. \quad (4.41)$$

This system is uniquely solvable if and only if

$$\ker(\tilde{A}^E) \cap \ker(\tilde{C}^E) = \{0\}, \quad (4.42)$$

but unisolvency of the dofs together with $\mathbb{P}_\ell(E) \subset \tilde{V}^E$ ensure that $\ker \tilde{A}^E = \{0\}$. Thus, the value projection of the nodal basis functions is uniquely solvable for as long as $\tilde{C}^E$ has full row rank.

4.1.3 Gradient projection

Definition 4.16 (Gradient projection [\[10\]](#)). For an element $E \in \mathcal{T}_h$, define the gradient projection Π_1^E into $\text{span} \mathcal{B}_1^E$ of any $v_h \in \tilde{V}^E$ as satisfying

$$\int_E \Pi_1^E [v_h] \cdot q \, dx = - \int_E \Pi_0^E [v_h] \nabla \cdot q \, dx + \sum_{e \in \mathcal{S}_h(E)} \int_e \Pi_0^e [v_h] (n \cdot q) \, ds \quad (4.43)$$

for all $q \in \mathcal{B}_1^E$, where n denotes the outward normal to edge e .

Lemma 4.17 (Well-posedness of the gradient projection). *Suppose that*

1. *For each $v_h \in \tilde{V}^E$ and edge $e \in \mathcal{S}_h(E)$, the value projections $\Pi_0^E[v_h]$ and $\Pi_0^e[v_h]$ are admissible.*
2. *$\text{span}\mathcal{B}_1^E \subset [L^2(E)]^2$, and for every $q \in \text{span}\mathcal{B}_1^E$, $\nabla \cdot q \in L^2(E)$ and $q \cdot n|_e \in L^2(e)$ on each edge $e \in \mathcal{S}_h(E)$.*

Then, for every $v_h \in \tilde{V}^E$, the variational problem (4.43) admits a unique solution $\Pi_1^E[v_h] \in \text{span}(\mathcal{B}_1^E)$, which is linear in v_h .

Proof. Let $W_1^E := \text{span}\mathcal{B}_1^E$, and let $(\cdot, \cdot)$ denote the L^2 inner-product on W_1^E . For a fixed $v_h \in \tilde{V}^E$, define the linear functional $l_{v_h} \in (W_1^E)'$ by

$$l_{v_h}(q) := - \int_E \Pi_0^E[v_h] \nabla \cdot q \, dx + \sum_{e \in \mathcal{S}_h(E)} \int_e \Pi_0^e[v_h] (n \cdot q) \, ds. \quad (4.44)$$

By the Riesz representation theorem on the finite-dimensional inner product space $(W_1^E, (\cdot, \cdot))$, there exists a unique $g \in W_1^E$ such that

$$(g, q) = l_{v_h}(q) \quad \forall q \in W_1^E. \quad (4.45)$$

This g is exactly the gradient projection $\Pi_1^E[v_h]$. Linearity in v_h follows from linearity of both Π_0^E and the integrals. $\square$

Proposition 4.18 (Exactness on $\nabla \text{span}\mathcal{B}_0^E$). *Suppose that the gradient projection is well-posed in the sense of Lemma 4.17, $v_h \in \text{span}\mathcal{B}_0^E$, and $\nabla v_h \in \text{span}\mathcal{B}_1^E$. Then*

$$\Pi_1^E[v_h] = \nabla v_h. \quad (4.46)$$

Proof. By Lemma 4.13, $\Pi_0^E[v_h] = v_h$, and $\Pi_0^e[v_h] = v_h$ for each $e \in \mathcal{S}_h(E)$. Then for any $q \in \text{span}\mathcal{B}_1^E$, integration by parts gives

$$\int_E \nabla v_h \cdot q \, dx = - \int_E v_h \nabla \cdot q \, dx + \sum_{e \in \mathcal{S}_h(E)} \int_e v_h (n \cdot q) \, ds \quad (4.47)$$

$$= - \int_E \Pi_0^E[v_h] \nabla \cdot q \, dx + \sum_{e \in \mathcal{S}_h(E)} \int_e \Pi_0^e[v_h] (n \cdot q) \, ds \quad (4.48)$$

which is the variational definition for $\Pi_1^E[v_h]$. Uniqueness follows from Lemma 4.17. $\square$

4.2 Mapping theory

The projections (4.9) and (4.43) are defined on each physical element, but solving the associated constrained least-squares problem to obtain the value projection on each element can be computationally intensive, potentially increasing the start-up time of an implementation too. This section aims to transport the VEM tuple and projection operators from a fixed reference element to a generic physical element, which mirrors the finite-element study of mapping the nodal basis functions and function space, but with the added constraint that these basis functions are considered to be virtual. The resulting projection operators may not coincide with the native CLS projection operator Π_0^E on the physical element, so we must show that the mapped projections retain key properties such as admissibility.

4.2.1 Reconstructing the VEM tuple

As for finite elements, let $\hat{E}$ be a fixed reference triangle and let $F : \hat{E} \rightarrow E$ be an affine mapping to a generic physical element $E \in \mathcal{T}_h$. We will simplify notation in this two-element case, writing the VEM tuple on the reference element as

$$(\hat{E}, \hat{\mathcal{B}}_0, \hat{\Lambda}, \hat{\mathcal{C}}_0, \hat{\mathcal{B}}_1). \quad (4.49)$$

For finite elements, we began by identifying that $F^*(\hat{P}) = P$, and then established a map $\Lambda = V^{-1}F_{-*}(\hat{\Lambda})$, allowing us to compute a map from the reference nodal basis functions to the physical element. For virtual elements, mapping the function space amounts to mapping the projection basis sets, and mapping the dofs is identical. Finally, we must also map the constraint set for the value projection.

- Define the value projection basis by pull-back, since polynomial spaces are preserved by pull-back. That is,

$$\mathcal{B}_0 = F^*(\hat{\mathcal{B}}_0) := \left\{ F^*(\hat{m}) : \hat{m} \in \hat{\mathcal{B}}_0 \right\}. \quad (4.50a)$$

- Define the gradient projection basis by the contravariant Piola map, because the gradient projection is defined through divergence and normal-flux pairings, and Piola maps preserve these pairings under the reference map (see [Proposition 2.25](#)).

$$\mathcal{B}_1 = \mathcal{F}^{\text{div}}(\hat{\mathcal{B}}_1) := \left\{ \mathcal{F}^{\text{div}}(\hat{m}) : \hat{m} \in \hat{\mathcal{B}}_1 \right\}. \quad (4.50b)$$

- Treating $(\hat{E}, \tilde{V}^{\hat{E}}, \hat{\Lambda})$ as a finite element, we can use [Proposition 3.10](#) and [Theorem 3.11](#) to compute invertible matrix V satisfying $VF_*(\Lambda) = \hat{\Lambda}$ from the dofs. Then we have

$$\Lambda = V^{-1}F_{-*}(\hat{\Lambda}) \quad (4.50c)$$

as before.

- Since we have assumed that the constraints are of point-evaluation type or moment type, by [Lemma 2.21](#), the reference constraint set can be mapped by inverse push-forward:

$$\mathcal{C}_0 = F_{-*}(\hat{\mathcal{C}}_0) := \left\{ F_{-*}(\hat{\mathcal{C}}) : \hat{\mathcal{C}} \in \hat{\mathcal{C}}_0 \right\}. \quad (4.50d)$$

For each edge $\hat{e} \in \mathcal{S}_h(\hat{E})$, let $e = F(\hat{e}) \in \mathcal{S}_h(E)$ denote its image. In addition to the element tuple on $\hat{E}$, we also transport an edge tuple $(\hat{e}, \hat{\mathcal{B}}_0^{\hat{e}}, \hat{\Lambda}^{\hat{e}}, \hat{\mathcal{C}}_0^{\hat{e}})$ to the physical edge tuple $(e, \mathcal{B}_0^e, \Lambda^e, \mathcal{C}_0^e)$. As before, we define $\mathcal{B}_0^e = F^*(\hat{\mathcal{B}}_0^{\hat{e}})$ and $\mathcal{C}_0^e = F_{-*}(\hat{\mathcal{C}}_0^{\hat{e}})$, where F is understood as its restriction $F|_{\hat{e}} : \hat{e} \rightarrow e$. Λ^e is recovered as the subset of the physical dofs depending only on the trace on e , in the sense of [Remark 4.5](#), yielding a second dof map matrix $V^{\hat{e}}$.

4.2.2 Mapping the value projection

We now seek to construct an admissible value projection $\hat{\Pi}_0^E$ on E from $\Pi_0^{\hat{E}}$ and $F : \hat{E} \rightarrow E$, acting as a surrogate to the native CLS value projection Π_0^E on E .

First, we give the following proposition for the case when the mapped dof matrix V from (4.50c) is orthogonal.

Proposition 4.19 (Value projection under pull-back). *If V is orthogonal, the CLS value projection on the physical element E coincides with the pull-back of the reference CLS value projection. That is,*

$$F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) = \Pi_0^E[F^*(\hat{v}_h)], \quad \forall \hat{v}_h \in \tilde{V}^E. \quad (4.51)$$

For each edge $\hat{e} \in \mathcal{S}_h(\hat{E})$, let $e = F(\hat{e}) \in \mathcal{S}_h(E)$ denote its image. If $V^{\hat{e}}$ is orthogonal, we have

$$F^*(\Pi_0^e[\hat{v}_h]) = \Pi_0^e[F^*(\hat{v}_h)], \quad \forall \hat{v}_h \in \tilde{V}^E. \quad (4.52)$$

Proof. We prove the statement for the element value projection, and the edge projection case is identical.

Set $v_h := F^*(\hat{v}_h)$. Noting that $V^T V = I$, observe that for $\hat{p} \in \text{span} \hat{\mathcal{B}}_0$ with $p := F^*(\hat{p})$,

$$\sum_{\hat{\lambda} \in \hat{\Lambda}} \hat{\lambda}(\hat{p} - \hat{v}_h)^2 = \left\| \hat{\Lambda}(\hat{p} - \hat{v}_h) \right\|_2^2 \quad (4.53)$$

$$= \|V F_*(\Lambda)(\hat{p} - \hat{v}_h)\|_2^2 \quad (4.54)$$

$$= (F_*(\Lambda)(\hat{p} - \hat{v}_h))^T V^T V F_*(\Lambda)(\hat{p} - \hat{v}_h) \quad (4.55)$$

$$= (F_*(\Lambda)(\hat{p} - \hat{v}_h))^T F_*(\Lambda)(\hat{p} - \hat{v}_h) \quad (4.56)$$

$$= \|F_*(\Lambda)(\hat{p} - \hat{v}_h)\|_2^2 \quad (4.57)$$

$$= \|\Lambda(p - v_h)\|_2^2 \quad (4.58)$$

$$= \sum_{\lambda \in \Lambda} \lambda(p - v_h)^2, \quad (4.59)$$

so if $\Pi_0^{\hat{E}}[\hat{v}_h] = \hat{p}$ satisfies the least squares objective on $\hat{E}$, then $p = F^*(\Pi_0^{\hat{E}}[\hat{v}_h])$ satisfies the least squares objective on E . Checking the constraints

$$0 = \hat{\mathcal{C}}(\Pi_0^{\hat{E}}[\hat{v}_h] - \hat{v}_h) = F_*(\mathcal{C})(\Pi_0^{\hat{E}}[\hat{v}_h] - \hat{v}_h) = \mathcal{C}(F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) - v_h), \quad (4.60)$$

since $\mathcal{C} = F_{-*}(\hat{\mathcal{C}})$ for all $\mathcal{C} \in \mathcal{C}_0$.

□

We define the (inverse) push-forward for projectors in the natural way, by

$$F_{-*}(\Pi_0^{\hat{E}})[v_h] := \Pi_0^{\hat{E}}[F^{-*}(v_h)]. \quad (4.61)$$

Then, even if the dof maps are not orthogonal, we shall define the mapped reference projections on E by push-forward.

Definition 4.20 (Surrogate value projection on E). Define the surrogate value projection on E by

$$\hat{\Pi}_0^E := F^*(F_{-*}(\Pi_0^{\hat{E}})). \quad (4.62)$$

That is, for $v_h \in \tilde{V}^E$,

$$\hat{\Pi}_0^E[v_h] := F^*(F_{-*}(\Pi_0^{\hat{E}}[v_h])) := \Pi_0^{\hat{E}}[v_h \circ F] \circ F^{-1}. \quad (4.63)$$

For each $\hat{e} \in \mathcal{S}_h(\hat{E})$ with image $e = F(\hat{e})$, define the surrogate edge value projection on e by

$$\hat{\Pi}_0^e := F^*(F_{-*}(\Pi_0^e)), \quad (4.64)$$

where F is understood as the restricted map $F|_{\hat{e}} : \hat{e} \rightarrow e$.

If V is not orthogonal, this projection need not be the CLS value projection on E ; however, we can show that it preserves the desirable properties of a value projection.

Theorem 4.21. *If $\Pi_0^{\hat{E}}$ is admissible on $\hat{E}$, then the projection $\hat{\Pi}_0^E$ is an admissible value projection on E .*

Proof. We check the three conditions of Definition 4.6.

1. Writing $v_h = \Lambda(v_h) \cdot \Phi$ in the physical nodal basis, we have that $\Phi = MF^*(\hat{\Phi})$, so $v_h = \Lambda(v_h) \cdot MF^*(\hat{\Phi})$. Hence,

$$\hat{\Pi}_0^E[v_h] = F^*(\Pi_0^{\hat{E}}[F^{-*}(\Lambda(v_h) \cdot MF^*(\hat{\Phi}))]) \quad (4.65)$$

$$= F^*(\Pi_0^{\hat{E}}[\Lambda(v_h) \cdot M\hat{\Phi}]) \quad (4.66)$$

$$= \Lambda(v_h) \cdot MF^*(\Pi_0^{\hat{E}}[\hat{\Phi}]). \quad (4.67)$$

Since we can express $\hat{\Pi}_0^E$ as a linear combination of the physical dofs $\Lambda^E(v_h)$, it is computable from Λ^E in the sense of Definition 2.14.

2. We have by construction that $\text{span}\mathcal{B}_0 = F^*(\hat{\mathcal{B}}_0)$, so define $\hat{p} = F^{-*}(p)$. Since Π_0^E is admissible, $\Pi_0^{\hat{E}}[\hat{p}] = \hat{p}$. Hence,

$$\hat{\Pi}_0^E[p] = F^*(\Pi_0^{\hat{E}}[F^{-*}(p)]) = F^*(\Pi_0^{\hat{E}}[\hat{p}]) = F^*(\hat{p}) = p. \quad (4.68)$$

3. Finally,

$$\mathcal{C}(\hat{\Pi}_0^E[v_h] - v_h) = \mathcal{C}(F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) - F^*(\hat{v}_h)) \quad (4.69)$$

$$= F_*(\mathcal{C})(\Pi_0^{\hat{E}}[\hat{v}_h] - \hat{v}_h) \quad (4.70)$$

$$= \hat{\mathcal{C}}(\Pi_0^{\hat{E}}[\hat{v}_h] - \hat{v}_h) \quad (4.71)$$

$$= 0, \quad (4.72)$$

since $\mathcal{C} = F_{-*}(\hat{\mathcal{C}})$ for every $\mathcal{C} \in \mathcal{C}_0$.

□

Corollary 4.22 (Condition for admissibility of $\hat{\Pi}_0^E$). *Suppose that for every $\hat{v}_h \in \tilde{V}^{\hat{E}}$, $\mathcal{A}_{\hat{v}_h}^{\hat{E}}$ is non-empty. Then $\hat{\Pi}_0^E$ is an admissible value projection.*

Proof. By Theorem 4.14, $\Pi_0^{\hat{E}}$ is admissible if for every $\hat{v}_h \in \tilde{V}^{\hat{E}}$, $\mathcal{A}_{\hat{v}_h}^{\hat{E}}$ is non-empty, so Theorem 4.21 gives the result. $\square$

Lemma 4.23. *For each $\hat{e} \in \mathcal{S}_h(\hat{E})$ with image $e = F(\hat{e})$, if $\Pi_0^{\hat{e}}$ is admissible on $\hat{e}$, then the projection $\hat{\Pi}_0^e$ is an admissible edge value projection on e .*

Proof. The proof is identical to that of Theorem 4.21, with element quantities replaced with the corresponding edge quantities, and with F replaced with $F|_{\hat{e}}$. $\square$

4.2.3 Constructing the mapped value projection

The decomposition used to prove computability of $\hat{\Pi}_0^E$ in Theorem 4.21 will also serve as the method of construction in an implementation. Using the result

$$\hat{\Pi}_0^E[v_h] = \Lambda(v_h) \cdot MF^*(\Pi_0^{\hat{E}}[\hat{\Phi}]), \quad (4.73)$$

we can compute the surrogate value projection of a generic function by pulling back the projection of the reference nodal basis functions, which can be precomputed. This reduces the global computational cost of computing the value projection by moving from solving a constrained least-squares problem on each element to solving it on the reference element and applying a pull-back and a linear map to each element. The reference value projection $\Pi_0^{\hat{E}}$ can be constructed as in Section 4.1.2.

4.2.4 Mapping the gradient projection

We now seek an appropriate mapping of the gradient projection on the reference element to a general element E . Recall that for an affine F , J is constant, so Example 2.27 gave

$$\nabla F^*(\hat{v}) = \mathcal{F}^{\text{curl}}(\hat{\nabla} \hat{v}). \quad (4.74)$$

To preserve the exactness of the gradient projection that was established in Proposition 4.18, we must apply the covariant Piola map (multiply by J^{-T} after pull-back) to the gradient projection.

As in (4.61), the (inverse) push-forward for operators is defined by applying the (inverse) pull-back to the argument.

Definition 4.24 (Surrogate gradient projection on E). For an element $E \in \mathcal{T}_h$, define

$$\hat{\Pi}_1^E := \mathcal{F}^{\text{curl}}(F_{-*}(\Pi_1^{\hat{E}})). \quad (4.75)$$

First, we prove that this gradient projection satisfies the same variational identity (4.43), with the CLS value projection replaced by $\hat{\Pi}_0^E$. This makes reference-mapped analogues of Lemma 4.17 and Proposition 4.18 almost automatic.

Theorem 4.25. *The projection $\hat{\Pi}_1^E : \tilde{V}^E \rightarrow \text{span}(\mathcal{B}_1)$ satisfies the variational identity*

$$\int_E \hat{\Pi}_1^E[v_h] \cdot q \, dx = - \int_E \hat{\Pi}_0^E[v_h] \nabla \cdot q \, dx + \sum_{e \in \mathcal{S}_h(E)} \int_e \hat{\Pi}_0^e[v_h] (n \cdot q) \, ds \quad (4.76)$$

on E . If the element and edge dof map matrices V and V^e are orthogonal, $\hat{\Pi}_1^E$ coincides with Π_1^E .

Proof. Let $v_h = F^*(\hat{v}_h) \in \tilde{V}^E$. Recalling that $\det J > 0$, for all $q := \mathcal{F}^{\text{div}}(\hat{q}) \in \mathcal{B}_1$ we get

$$\int_E \hat{\Pi}_1^E[v_h] \cdot q \, dx = \int_E J^{-T} F^*(\Pi_1^{\hat{E}}[\hat{v}_h]) \cdot q \, dx \quad (4.77)$$

$$= \int_{\hat{E}} J^{-T} \Pi_1^{\hat{E}}[\hat{v}_h] \cdot F^{-*}(q) \det(J) \, d\hat{x} \quad (4.78)$$

$$= \int_{\hat{E}} J^{-T} \Pi_1^{\hat{E}}[\hat{v}_h] \cdot \left(\frac{1}{\det J} J \hat{q} \right) \det(J) \, d\hat{x} \quad (4.79)$$

$$= \int_{\hat{E}} \Pi_1^{\hat{E}}[\hat{v}_h] \cdot J^{-1} J \hat{q} \, d\hat{x} \quad (4.80)$$

$$= \int_{\hat{E}} \Pi_1^{\hat{E}}[\hat{v}_h] \cdot \hat{q} \, d\hat{x} \quad (4.81)$$

$$:= - \int_{\hat{E}} \Pi_0^{\hat{E}}[\hat{v}_h] \hat{\nabla} \cdot \hat{q} \, d\hat{x} + \sum_{\hat{e} \in \mathcal{S}_h(\hat{E})} \int_{\hat{e}} \Pi_0^{\hat{E}}[\hat{v}_h] \hat{n} \cdot \hat{q} \, d\hat{s}. \quad (4.82)$$

The chain rule gives that

$$\hat{\nabla} \cdot \hat{q} = \hat{\nabla} \cdot (\det(J) J^{-1} F^{-*}(q)) \quad (4.83)$$

$$= \det(J) F^{-*}(\text{tr}(J^{-1} J \nabla q)) \quad (4.84)$$

$$= \det(J) F^{-*}(\nabla \cdot q), \quad (4.85)$$

and hence

$$F^*(\hat{\nabla} \cdot \hat{q}) = \det(J) (\nabla \cdot q). \quad (4.86)$$

Transforming the volume term of (4.82) back to E gives

$$- \int_{\hat{E}} \Pi_0^{\hat{E}}[\hat{v}_h] \hat{\nabla} \cdot \hat{q} \, d\hat{x} = - \int_E F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) F^*(\hat{\nabla} \cdot \hat{q}) \frac{1}{\det J} \, dx \quad (4.87)$$

$$= - \int_E \hat{\Pi}_0^E[v_h] \det(J) \nabla \cdot q \frac{1}{\det J} \, dx \quad (4.88)$$

$$= - \int_E \hat{\Pi}_0^E[v_h] \nabla \cdot q \, dx, \quad (4.89)$$

where we used that $F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) = F^*(\Pi_0^{\hat{E}}[F^{-*}(v_h)]) = F^*(F_{-*}(\Pi_0^{\hat{E}}[v_h])) =: \hat{\Pi}_0^E[v_h]$.

By the normal-flux preservation property of the contravariant Piola transformation from [Proposition 2.25](#), normal integral moments are preserved, and so the boundary terms of (4.82) satisfy

$$\int_{\hat{e}} \Pi_0^{\hat{E}}[\hat{v}_h] \hat{n} \cdot \hat{q} \, d\hat{s} = \int_e F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) n \cdot \mathcal{F}^{\text{div}}(\hat{q}) \, ds, \quad (4.90)$$

$\forall \hat{e} \in \mathcal{S}_h(\hat{E})$, and $e = F(\hat{e})$. Since $q = \mathcal{F}^{\text{div}}(\hat{q})$, we have

$$\int_e F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) n \cdot \mathcal{F}^{\text{div}}(\hat{q}) \, ds = \int_e \hat{\Pi}_0^E[v_h] n \cdot q \, ds, \quad (4.91)$$

as required.

If V and V^e are orthogonal, then [Proposition 4.19](#) ensures $F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) = \Pi_0^E[v_h]$ and $F^*(\Pi_0^{\hat{E}}[\hat{v}_h]) = \Pi_0^E[v_h]$, while [Lemma 4.17](#) ensures uniqueness of the gradient projection, so $\hat{\Pi}_1^E[v_h] = \Pi_1^E[v_h]$. $\square$

Lemma 4.26 (Well-posedness of $\hat{\Pi}_1^E$). *Suppose that*

1. *For each $v_h \in \tilde{V}^E$ and edge $e \in \mathcal{S}_h(E)$, the value projections $\hat{\Pi}_0^E[v_h]$ and $\hat{\Pi}_0^e[v_h]$ are admissible.*
2. *$\text{span}\mathcal{B}_1^E \subset [L^2(E)]^2$, and for every $q \in \text{span}\mathcal{B}_1^E$, $\nabla \cdot q \in L^2(E)$ and $q \cdot n|_e \in L^2(e)$ on each edge $e \in \mathcal{S}_h(E)$.*

Then, for every $v_h \in \tilde{V}^E$, the variational problem (4.76) admits a unique solution $\hat{\Pi}_1^E[v_h] \in \text{span}(\mathcal{B}_1^E)$, which is linear in v_h .

Proof. The proof is identical to the proof of Lemma 4.17, with the value projection Π_0 replaced by $\hat{\Pi}_0$. $\square$

Proposition 4.27 (Exactness on $\nabla \text{span}\mathcal{B}_0^E$). *Suppose that $\hat{\Pi}_1^E$ is well-posed in the sense of Lemma 4.26, $v_h \in \text{span}\mathcal{B}_0^E$, and $\nabla v_h \in \text{span}\mathcal{B}_1^E$. Then*

$$\hat{\Pi}_1^E[v_h] = \nabla v_h. \quad (4.92)$$

Proof. The proof is identical to the proof of Proposition 4.18, with Theorem 4.21 giving exactness of $\hat{\Pi}_0^E$ and $\hat{\Pi}_0^e$, and uniqueness following from Lemma 4.26. $\square$

5 Examples

We will now demonstrate the theory we have developed with some different virtual element spaces. Throughout Section 4, we assumed several properties of a VEM tuple over a VEM space, and we have asserted premises under which the projections we require on such a space are suitable. For convenience, a summary of conditions has been added to the appendix as Section 9.1.2, which we will refer to frequently. We exhibit our VEM spaces on a model problem: for some $f \in L^2(\Omega)$, we consider the Poisson problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \quad (5.1)$$

Using the common notation for the Sobolev spaces $H^1(\Omega)$ and $H_0^1(\Omega)$, letting $(\cdot, \cdot)$ denote the L^2 -inner product, and defining $a(u, v) := (\nabla u, \nabla v)$, the weak formulation of (5.1) is to find $u \in H_0^1(\Omega)$ such that

$$a(u, v) = (f, v), \quad \forall v \in H_0^1(\Omega). \quad (5.2)$$

By the Lax-Milgram lemma, this problem admits a unique solution.

5.1 A Lagrange-Type Virtual Element Space

5.1.1 Virtual element space

The virtual shape function space and dofs in this section are from the H^1 -conforming case in [7], with the constraints and basis sets chosen to support the swap to a CLS value projection in [10].

Let $\hat{E}$ be a triangular reference element and $k \geq 1$ be an integer. Assume that our equation admits a scalar-valued solution. Then, define the local shape function space to be

$$\tilde{V}^{\hat{E}} = \{v \in H^1(\hat{E}) : \Delta v|_{\hat{E}} \in \mathbb{P}_{k-2}(\hat{E}), v|_{\partial\hat{E}} \in \mathbb{B}_k(\partial\hat{E})\}, \quad (5.3)$$

where we take the convention that $\mathbb{P}_{-1}(\Omega) = \{0\}$.

Remark 5.1. The condition $\Delta v|_{\hat{E}} \in \mathbb{P}_{k-2}(\hat{E})$ is to be understood in the weak sense. No pointwise second derivatives are required, and functions in $\tilde{V}^{\hat{E}}$ need not belong to $C^2(\hat{E})$.

Define the reference VEM tuple $(\hat{E}, \hat{\mathcal{B}}_0, \hat{\Lambda}, \hat{\mathcal{C}}_0, \hat{\mathcal{B}}_1)$ on $\hat{E}$, consisting of

- Value projection basis

$$\hat{\mathcal{B}}_0 = \mathcal{M}_k(\hat{E}). \quad (5.4)$$

- Degrees of freedom $\hat{\Lambda}$ containing:

1. The values at the vertices

$$\delta_{v_i} \hat{v}_h, \quad \forall i \in \{1, 2, 3\}. \quad (5.5a)$$

2. For $k \geq 2$, the edge moments

$$\frac{1}{|\hat{e}_i|} \int_{\hat{e}_i} \hat{q} \, d\hat{s} \quad \forall \hat{q} \in \mathcal{M}_{k-2}(\hat{e}_i), \quad i \in \{1, 2, 3\}. \quad (5.5b)$$

3. For $k \geq 2$, the interior moments

$$\frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{v}_h \hat{q} \, d\hat{x} \quad \forall \hat{q} \in \mathcal{M}_{k-2}(\hat{E}). \quad (5.5c)$$

- Constraint set $\hat{\mathcal{C}}_0$ containing

$$\hat{\mathcal{C}}(\hat{v}_h) = \frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{v}_h \hat{q} \, d\hat{x} \quad \forall \hat{q} \in \mathcal{M}_{k-2}(\hat{E}). \quad (5.6)$$

- Gradient projection basis set

$$\hat{\mathcal{B}}_1 = [\mathcal{M}_{k-1}(\hat{E})]^2. \quad (5.7)$$

Assumptions 2,3 and 5 of [Section 9.1.2](#) are clear, and since the constraints are a subset of the dofs, so is 4. Thus, it remains to check unisolvency of the dofs for $\tilde{V}^{\hat{E}}$.

Lemma 5.2. *The degrees of freedom (5.5) are unisolvent for the space $\tilde{V}^{\hat{E}}$.*

Proof. Firstly,

$$\dim \tilde{V}^{\hat{E}} = \dim \mathbb{P}_{k-2}(\hat{E}) + \dim \mathbb{B}_k(\partial\hat{E}) = \frac{1}{2}k(k-1) + 3k = |\hat{\Lambda}| \quad (5.8)$$

We will prove that if a function $\hat{v}_h \in \tilde{V}^{\hat{E}}$ satisfies $\hat{\lambda}(\hat{v}_h) = 0$ for all $\hat{\lambda} \in \hat{\Lambda}$, then $v_h \equiv 0$. The degrees of freedom (5.5a) - (5.5b) uniquely determine a degree k polynomial on each edge, which must be the zero polynomial since all dofs vanish at $\hat{v}_h$, so $\hat{v}_h|_{\partial\hat{E}} = 0$. If $k \leq 2$, then the edge moments and vertex evaluations fully determine $\hat{v}_h$, so we are done. Otherwise, (5.5c) implies that

$$\int_{\hat{E}} \hat{v}_h \hat{p} \, d\hat{x} = 0 \quad \forall \hat{p} \in \mathbb{P}_{k-2}(\hat{E}). \quad (5.9)$$

Since we have that $\hat{\Delta}\hat{v}_h \in \mathbb{P}_{k-2}(\hat{E})$ for all $\hat{v}_h \in \tilde{V}^{\hat{E}}$,

$$\int_{\hat{E}} \hat{v}_h \hat{\Delta}\hat{v}_h \, d\hat{x} = 0. \quad (5.10)$$

Finally, by Green's identity,

$$\int_{\hat{E}} |\hat{\nabla}\hat{v}_h|^2 \, d\hat{x} = - \int_{\hat{E}} \hat{v}_h \hat{\Delta}\hat{v}_h \, d\hat{x} + \int_{\partial\hat{E}} \hat{v}_h (\hat{\nabla}\hat{v}_h \cdot \hat{n}) \, d\hat{s} = 0. \quad (5.11)$$

Thus $\hat{v}_h$ is constant, so equal to 0 on all of $\hat{E}$. $\square$

Continuing to check the conditions of Section 9.1.2, we show that the feasible sets are non-empty.

Corollary 5.3. *For every $\hat{v}_h$ in $\tilde{V}^{\hat{E}}$,*

- *the feasible set $\mathcal{A}_{\hat{v}_h}^{\hat{E}}$ is non-empty, and*
- *the feasible set $\mathcal{A}_{\hat{v}_h}^{\hat{e}}$ is non-empty for every $\hat{e} \in \mathcal{S}_h(\hat{E})$.*

Proof. The element feasible set for this space is

$$\mathcal{A}_{\hat{v}_h} := \left\{ \hat{p} \in \mathbb{P}_k(\hat{E}) : \frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{p} \hat{q} \, d\hat{x} = \frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{v}_h \hat{q} \, d\hat{x} \quad \forall \hat{q} \in \mathcal{M}_{k-2}(\hat{E}) \right\}. \quad (5.12)$$

The system

$$\frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{p} \hat{q} \, d\hat{x} = \frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{v}_h \hat{q} \, d\hat{x} \quad \forall \hat{q} \in \mathcal{M}_{k-2}(\hat{E}), \quad (5.13)$$

defines the L^2 -projection of $\hat{v}_h$ onto $\mathbb{P}_{k-2}(\hat{E})$. It is well-posed because the L^2 mass matrix on $\mathbb{P}_{k-2}(\hat{E})$ is symmetric positive definite. Thus it admits a unique solution $\hat{p}_0 \in \mathcal{A}_{\hat{v}_h}$.

Since we have no edge constraints, $\mathcal{A}_{\hat{v}_h}^{\hat{e}} = \mathbb{B}_k(\hat{e}) \neq \emptyset$. $\square$

Therefore, the CLS value projections are admissible on the reference element. It holds that $\text{span}\mathcal{B}_1^{\hat{E}} = [\mathbb{P}_{k-1}(\hat{E})]^2 \subset [L^2(\hat{E})]^2$, and for every $\hat{q} \in [\mathbb{P}_{k-1}(\hat{E})]^2$ and $i \in \{1, 2, 3\}$, we have $\hat{\nabla} \cdot \hat{q}$, $\hat{q} \cdot \hat{n}_i \in L^2(\hat{E})$. We conclude by Section 9.1.2 that the gradient projection $\Pi_1^{\hat{E}}$ is well-posed and exact on $\nabla \text{span}\mathbb{P}_k(\hat{E})$.

5.1.2 Mapping

Let $F : \hat{E} \rightarrow E$ be an affine mapping. The VEM tuple on E from (4.50) only requires us to construct the matrix V satisfying

$$VF_*(\Lambda) = \hat{\Lambda}. \quad (5.14)$$

Lemma 2.21 established that point evaluations and integral moments push forward trivially, so we conclude that $V = I$ for all orders k , matching the finite element case Example 3.13. As a consequence, $\hat{\Pi}_0^E$ from Definition 4.20 coincides with the CLS value projection on E , for all orders k , pictured in Figure 6.

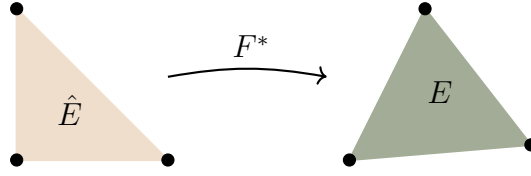


Figure 6: The pull-back operation preserves the value projection, pictured on the lowest order Lagrange element (see [2]).

5.2 A Hermite-Type Virtual Element Space

5.2.1 Virtual element space

The virtual shape function space and dofs in this section are from [18], with the constraints and basis sets chosen to support the swap to a CLS value projection.

Again, let $\hat{E}$ be a triangular reference element and $k \geq 3$ be an integer. The shape function space is identical to (5.3):

$$\tilde{V}^{\hat{E}} = \{v \in H^1(\hat{E}) : \Delta v|_{\hat{E}} \in \mathbb{P}_{k-2}(\hat{E}), v|_{\partial\hat{E}} \in \mathbb{B}_k(\partial\hat{E})\}. \quad (5.15)$$

We now include gradient degrees of freedom when defining the reference virtual element tuple. The basis sets and constraints are unchanged from Section 5.1, so we only define the degrees of freedom $\hat{\Lambda}$, which contain:

1. The values at the vertices

$$\delta_{v_i} \hat{v}_h, \quad i \in \{1, 2, 3\}. \quad (5.16a)$$

2. The gradient values at the vertices

$$h_{v_i} \nabla_{v_i} \hat{v}_h, \quad \forall i \in \{1, 2, 3\}, \quad (5.16b)$$

where h_{v_i} is the characteristic length associated with the vertex v_i (the average length of an edge attached to v_i).

3. The edge moments

$$\frac{1}{|\hat{e}_i|} \int_{\hat{e}_i} \hat{v}_h \hat{q} \, d\hat{s} \quad \forall \hat{q} \in \mathcal{M}_{k-4}(\hat{E}), \quad \forall i \in \{1, 2, 3\}. \quad (5.16c)$$

4. The interior moments

$$\frac{1}{|\hat{E}|} \int_{\hat{E}} \hat{v}_h \hat{q} \, d\hat{x} \quad \forall \hat{q} \in \mathcal{M}_{k-2}(\hat{E}). \quad (5.16d)$$

Once again, only assumption 4 of [Section 9.1.2](#) requires checking.

Lemma 5.4. *The degrees of freedom (5.16) are unisolvent for the space $\tilde{V}^{\hat{E}}$.*

Proof. We follow the same formula as in [Lemma 5.2](#). Firstly,

$$\dim \tilde{V}^{\hat{E}} = \dim \mathbb{P}_{k-2}(\hat{E}) + \dim \mathbb{B}_k(\partial \hat{E}) = \frac{1}{2}(k-1)(k-2) + 3k = |\hat{\Lambda}|. \quad (5.17)$$

Next, if $\hat{v}_h \in \tilde{V}^{\hat{E}}$ satisfies $\hat{\lambda}(\hat{v}_h) = 0$ for all $\hat{\lambda} \in \hat{\Lambda}$, then the degrees of freedom (5.16a) - (5.16c) uniquely determine a degree k polynomial on each edge, which must be the zero polynomial since all dofs vanish at $\hat{v}_h$, so $\hat{v}_h|_{\partial \hat{E}} = 0$. Since we have that $\hat{\Delta} \hat{v}_h \in \mathbb{P}_{k-2}(\hat{E})$ for all $\hat{v}_h \in \tilde{V}^{\hat{E}}$, (5.16d) implies that

$$\int_{\hat{E}} \hat{v}_h \hat{\Delta} \hat{v}_h \, d\hat{x} = 0. \quad (5.18)$$

Finally, by Green's identity,

$$\int_{\hat{E}} |\hat{\nabla} \hat{v}_h|^2 \, d\hat{x} = - \int_{\hat{E}} \hat{v}_h \hat{\Delta} \hat{v}_h \, d\hat{x} + \int_{\partial \hat{E}} \hat{v}_h (\hat{\nabla} \hat{v}_h \cdot \mathbf{n}) \, d\hat{s} = 0. \quad (5.19)$$

Thus $\hat{v}_h$ is constant, so equal to 0 on all of $\hat{E}$, as required. $\square$

Proceeding as in [Corollary 5.3](#) and the discussion following it, we find that the CLS value projections are admissible and the gradient projection is well-posed and exact on $\nabla \mathbb{P}_k(\hat{E})$.

5.2.2 Mapping

For a physical element E and an affine map $F : \hat{E} \rightarrow E$, we must find V so that $VF_*(\Lambda) = \hat{\Lambda}$. The Lagrange-type virtual element established that the point evaluation and moment dofs map trivially when pushed forward, so only the gradient nodes contribute to V , as in (3.25). Thus, V is block diagonal, with the only non-identity blocks being

$$\frac{h_{\hat{v}_i}}{h_{v_i}} J^T, \quad (5.20)$$

corresponding to the gradient node at vertex v_i .

By [Section 9.1.2](#), we have that $\hat{\Pi}_0^E$ and $\hat{\Pi}_0^e$ are admissible value projections for each $e \in \mathcal{S}_h(E)$, and $\hat{\Pi}_1^E$ is well-defined on E and exact on $\nabla \text{span} \mathbb{P}_k(E)$. Unlike the Lagrange-type virtual element, since V is not necessarily orthogonal, these projections may not coincide with the native CLS value projection and gradient projection on E . In the next section, we will see that they behave as expected, converging correctly in a test case.

6 Implementation and Results

To conform to page requirements, figures referenced in this section are contained in the appendix. Alongside this document, a GitHub repository has been developed to implement the ideas presented, available at github.com/joshuadendy/Virtual-Element-Method [11]. This section uses the demonstration `demo/run_poisson.py`, and results can be reproduced using the arguments specified by [Code snippet 1](#).

Using this script, we will solve the Poisson equation (5.1) with

$$f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y), \quad (6.1)$$

on $\Omega = [0, 1]^2$, using Dirichlet boundary conditions $u|_{\partial\Omega} = 0$. This equation has analytical solution

$$u(x, y) = \sin(\pi x) \sin(\pi y), \quad (6.2)$$

which we will use to measure our model's performance.

[Figure 7](#) shows the true solution (6.2) imposed on a 32-triangle mesh. Leaving the implementation details to the repository, we will run the model 5 times, refining the mesh by increasing the number of elements by a factor of 4 each time.

We test a range of elements from the examples, denoting classic VEM spaces in which the CLS value projection (4.9) and the resultant gradient projection (4.43) are computed on each element as physical. Reference mapped spaces using [Section 4.2.3](#) to compute the surrogate value projection and (4.76) for the surrogate gradient projection on each element as mapped. We use the $k = 1$ and $k = 2$ Lagrange-type VEM spaces from [Section 5.1](#), and the $k = 3$ and $k = 4$ Hermite-type VEM spaces from [Section 5.2](#).

[Table 2](#) gives the diameter, number of elements, and dof counts for each space on the range of meshes. [Figure 8](#) shows the estimated order of convergence (EOC) for the L^2 -norm of the error between the true solution and the approximated solution u_h , given by

$$\|u - u_h\|_{L^2} = \left(\sum_{E \in \mathcal{T}_h} \int_E |u - u_h|^2 dx \right)^{\frac{1}{2}}. \quad (6.3)$$

[Figure 9](#) shows the H^1 -seminorm of the error, given by

$$|u - u_h|_{H^1} = \left(\sum_{E \in \mathcal{T}_h} \int_E |\nabla(u - u_h)|^2 dx \right)^{\frac{1}{2}}. \quad (6.4)$$

These plots show that, experimentally, the surrogate projections converge at the same rate as the CLS counterparts - roughly order k in the H^1 -seminorm and order $k + 1$ in the L^2 -norm. Although no formal timing was carried out, the mapped-space implementation was consistently observed to run faster in practice. [Table 3](#) shows the total runtime for each space on each mesh refinement, showing that, for this experiment, the mapped spaces performed significantly faster with identical results. [Figure 10](#) shows the absolute errors for the mapped spaces on the 32-element mesh. [Table 4](#) and [Table 5](#) contain the difference in the two errors between the mapped and physical spaces, demonstrating that while the errors are very close, there may be a dependency on h , especially for the difference in L^2 -error of the $k = 4$ Hermite space.

Due to time constraints, this experiment used a square mesh of identical triangles, which neither demonstrates the full capabilities of VEM nor fully tests the reference mapping functionality. Future releases will extend to more irregular grids and possibly non-triangular elements.

7 Conclusion and Outlook

This project has considered the role of reference maps in constructing virtual element spaces on triangular meshes. Motivated by the use of reference elements in the finite element method, the aim has been to investigate a similar approach for virtual element methods in the constrained least squares framework. The main outcome in [Section 4.2](#) is (4.62), an admissible value projection, which is a viable alternative to the native CLS value projection. It was then shown that the gradient projection, computed from the reference-mapped value projection via [Definition 4.24](#), is well-posed and exact on the correct class of functions to make it a suitable replacement for the gradient projection (4.43). Construction of these projections was illustrated for both Lagrange-type and Hermite-type VEM spaces in [Section 5](#). The numerical experiments in [Section 6](#) showed that the mapped projections exhibited the same convergence behaviour as their native CLS counterparts, with neither admissible projection being obviously more accurate.

There are several natural next steps for this project, such as extending beyond triangles to more general polygonal grids, exploring dimensions higher than two, and implementing higher-order spaces, such as the $k = 5$ Hermite-type VEM in [\[11\]](#). It would also be of interest to study vector-valued virtual element spaces, particularly those which require Piola-type transformations [\[3\]](#). Finally, [\[10, section 4.2\]](#) uses the Hessian projection, which we have not mentioned explicitly, to produce an H^2 -conforming VEM space. The theory for this projection and the corresponding reference map should follow naturally from our framework, except that the space uses the constraint

$$\int_E \Delta \Pi_0^E[v_h] \, dx = \sum_{e \in \mathcal{S}_h(E)} \int_e \Pi_0^e[v_h], \quad (7.1)$$

which is not compatible with our framework due to the assumptions in [Definition 4.3](#). Following [Remark 9.1](#), expanding the range of possible constraints is possible, so implementing a reference-mapped Hessian projection is a suitable next priority.

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9 Appendix

9.1 Summaries

9.1.1 Summary of mapping notation

Map	Domain and Codomain	Definition
Affine map	$F : \hat{E} \rightarrow E$	$F(\hat{x}) = J\hat{x} + b$
Pull-back	$F^* : W_{\hat{E}} \rightarrow W_E$	$F^*(\hat{v}) = v \circ F^{-1}$
Push-forward	$F_* : W'_E \rightarrow W'_{\hat{E}}$	$F_*(Q) = Q \circ F^*$

Table 1: Mapping notation summary table.

9.1.2 Summary of VEM conditions

Throughout [Section 4](#), we assumed several properties of a VEM tuple over a VEM space, and we have asserted premises under which the projections we require on such a space are suitable. Here we present a summary of these conditions for reference in the examples.

Suppose $\hat{E}$ is a reference element. Let $\tilde{V}^{\hat{E}}$ be the shape function space on $\hat{E}$ and define the reference VEM tuple $(\hat{E}, \hat{\mathcal{B}}_0, \hat{\Lambda}, \hat{\mathcal{C}}_0, \hat{\mathcal{B}}_1)$. Similarly, define $(\hat{e}, \hat{\mathcal{B}}_0^{\hat{e}}, \hat{\Lambda}^{\hat{e}}, \hat{\mathcal{C}}_0^{\hat{e}})$ on each $\hat{e} \in \mathcal{S}_h(\hat{E})$. Suppose that

1. $\hat{\Lambda}$ is unisolvent on $\tilde{V}^{\hat{E}}$,
2. $\mathbb{P}_{\ell}(\hat{E}) \subset \tilde{V}^{\hat{E}}$ for some $\ell \geq 0$,
3. $\hat{\mathcal{B}}_0$ forms a basis of $\mathbb{P}_{\ell}(\hat{E})$,
4. $\hat{\mathcal{C}}_0$ is computable from $\hat{\Lambda}$, and
5. $\text{span} \hat{\mathcal{B}}_1 \subset \nabla \tilde{V}^{\hat{E}}$.

For every $\hat{v}_h$ in $\tilde{V}^{\hat{E}}$, if

- The feasible set $\mathcal{A}_{\hat{v}_h}^{\hat{E}}$ is non-empty then the CLS value projection $\Pi_0^{\hat{E}}$ is admissible.
- The feasible set $\mathcal{A}_{\hat{v}_h}^{\hat{e}}$ is non-empty then the CLS edge value projection $\Pi_0^{\hat{e}}$ is admissible.

If all of the above conditions hold, and in addition,

1. $\text{span} \hat{\mathcal{B}}_1 \subset L^2(\hat{E})$.

2. for every $q \in \text{span}\mathcal{B}_1^E$, $\nabla \cdot q \in L^2(E)$ and $q \cdot n|_e \in L^2(e)$ on each edge $e \in \mathcal{S}_h(E)$.

Then the gradient projection $\Pi_1^{\hat{E}}$ is well-defined on $\hat{E}$ and exact on $\nabla \text{span}\hat{\mathcal{B}}_0$.

Now let $\mathcal{T}_h$ be a mesh, $E \in \mathcal{T}_h$ be a physical element, and $F : \hat{E} \rightarrow E$ be an affine map. Then $\hat{\Pi}_0^E$ and $\hat{\Pi}_0^e$ are admissible value projections for each $e \in \mathcal{S}_h(E)$, and $\hat{\Pi}_1^E$ is well-defined on E and exact on $\nabla \text{span}\mathcal{B}_0^E$.

Remark 9.1 (Constraint assumptions). We have additionally assumed that the constraints are of point-evaluation type or moment type, which simplifies (4.50d) and the proof of Theorem 4.21. This condition can be relaxed by mapping the constraints under push-forward in a similar manner to the dofs in (4.50c) using the relation

$$F_{-*}(\hat{\mathcal{C}}) = M\hat{\mathcal{C}}(\hat{\Phi}) \cdot \Lambda, \quad (9.1)$$

although verifying admissibility of $\hat{\Pi}_0^E$ requires some extra work. In our examples, this assumption will always hold.

9.2 Numerical results figures

Code snippet 1: parameters used for Poisson equation demonstration.

```
# In the __main__ function of demo/run_poisson.py,
# replace the default arguments with:
run_poisson_demo(
    spaces=(
        LinearLagrangePhysicalVEMSpace,
        LinearLagrangeMappedVEMSpace,
        QuadraticLagrangePhysicalVEMSpace,
        QuadraticLagrangeMappedVEMSpace,
        CubicHermitePhysicalVEMSpace,
        CubicHermiteMappedVEMSpace,
        QuarticHermitePhysicalVEMSpace,
        QuarticHermiteMappedVEMSpace,
    ),
    refinements=5,
    plot=True,
    plot_true_solution=True,
    plot_error=True,
    compare_mapped=False,
    plot_eoc=True,
    nx0=4,
    ny0=4,
)
plt.show()
```

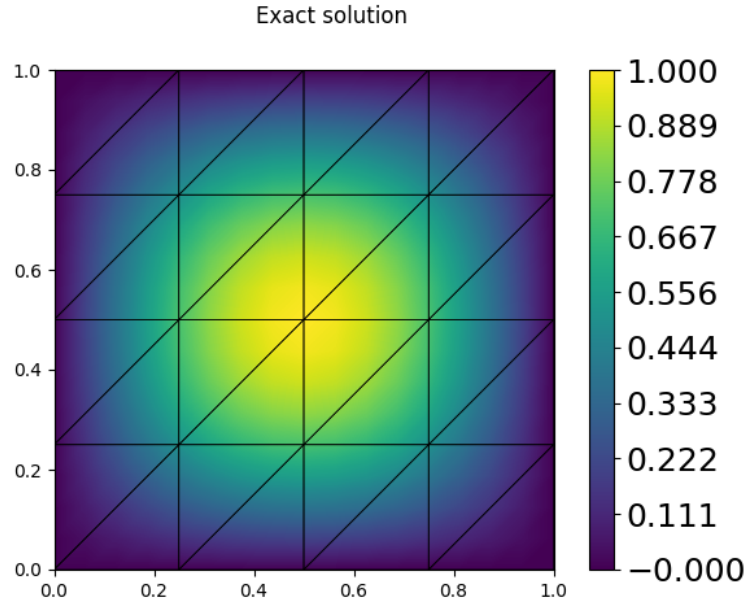


Figure 7: Initial mesh used and true solution to the Poisson equation (5.1). For this mesh, $h \approx 0.35$.

Refinement level	h	Element count	Number of dofs per space			
			Lagrange $k = 1$	Lagrange $k = 2$	Hermite $k = 3$	Hermite $k = 4$
0	0.35	32	25	113	171	323
1	0.18	128	81	417	627	1219
2	0.088	512	289	1601	2403	4739
3	0.044	2048	1089	6273	9411	18691
4	0.022	8192	4225	24833	37251	74243

Table 2: Mesh characteristics and number of dofs per space for each mesh refinement.

Space	0	1	2	3	4	Total
LLP	0.577	0.354	1.622	6.514	21.867	30.934
LLM	0.081	0.281	1.127	4.993	19.481	25.963
QLP	0.382	1.442	5.631	21.852	95.078	124.386
QLM	0.162	0.698	2.413	10.143	51.049	64.465
CHP	0.782	3.012	13.868	57.401	248.063	323.125
CHM	0.221	0.757	3.146	13.770	72.449	90.344
QHP	2.257	8.681	34.034	141.164	762.403	948.539
QHM	0.383	1.495	6.199	36.999	201.904	246.980

Table 3: Runtime per refinement level (seconds). Abbreviations: LLP = LinearLagrangePhysical, LLM = LinearLagrangeMapped, QLP = QuadraticLagrangePhysical, QLM = QuadraticLagrangeMapped, CHP = CubicHermitePhysical, CHM = CubicHermiteMapped, QHP = QuarticHermitePhysical, QHM = QuarticHermiteMapped.

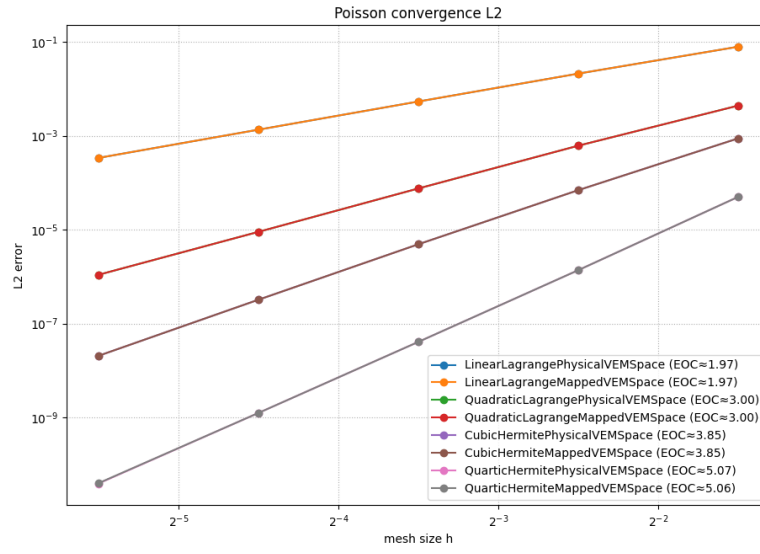


Figure 8: L^2 -norm EOCs for the various finite and virtual element spaces.

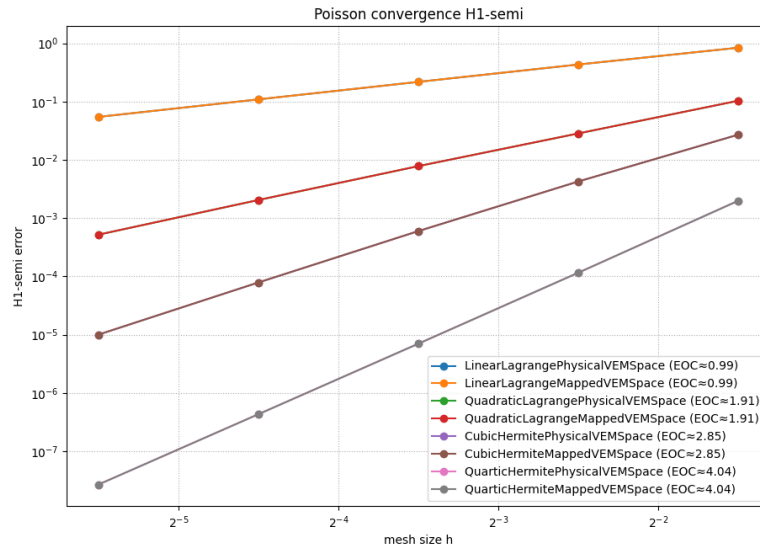


Figure 9: H^1 -seminorm EOCs for the various finite and virtual element spaces.

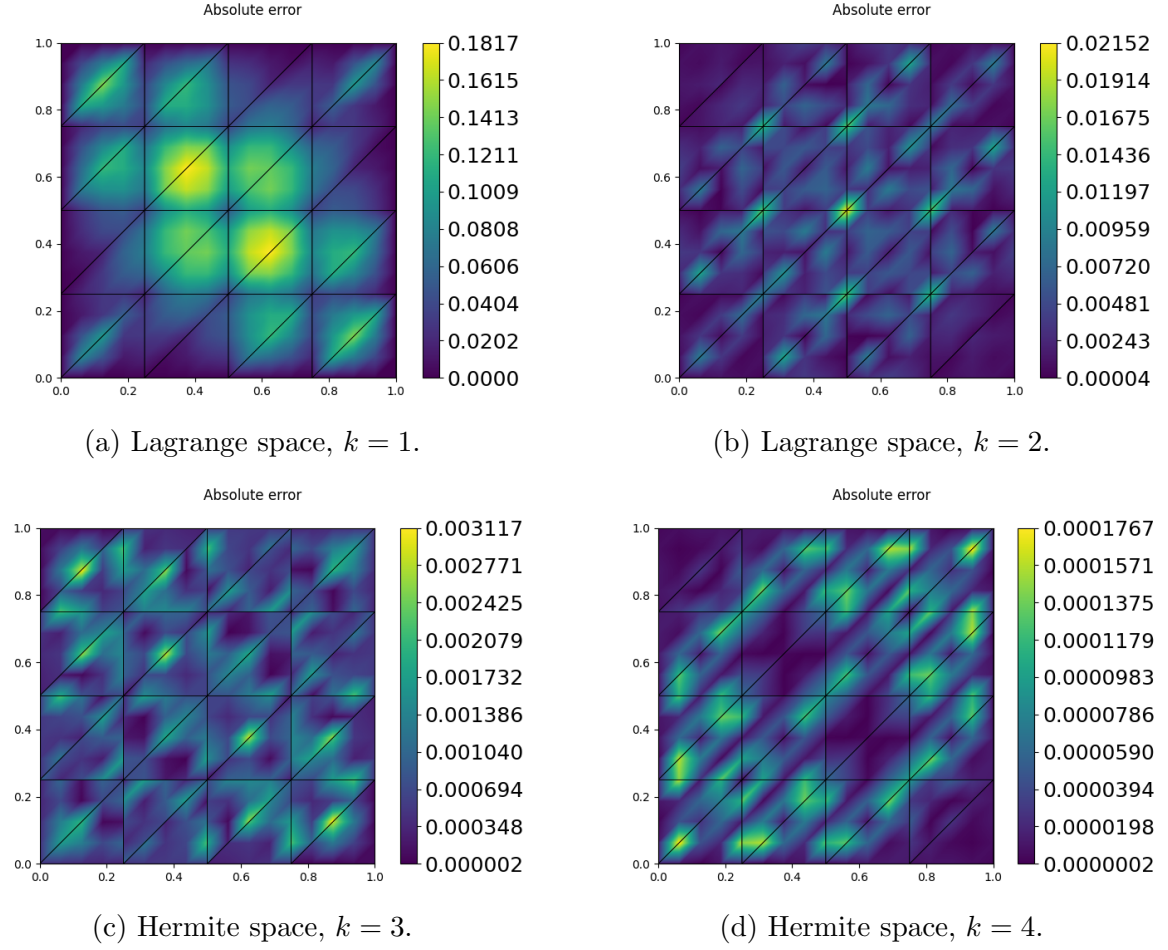


Figure 10: Absolute errors for the Lagrange and Hermite mapped VEM spaces. For readability, the colour bars have differing scales.

Refinement level	L^2 -error mapped vs physical			
	Lagrange $k = 1$	Lagrange $k = 2$	Hermite $k = 3$	Hermite $k = 4$
0	3.510×10^{-16}	3.221×10^{-13}	5.824×10^{-4}	2.148×10^{-6}
1	1.707×10^{-14}	3.659×10^{-12}	1.182×10^{-4}	5.298×10^{-6}
2	1.486×10^{-13}	1.018×10^{-10}	1.533×10^{-4}	5.913×10^{-6}
3	3.356×10^{-14}	2.349×10^{-9}	2.621×10^{-4}	5.108×10^{-6}
4	1.154×10^{-10}	4.139×10^{-8}	3.000×10^{-4}	1.711×10^{-2}

Table 4: $|e_{L^2}^m - e_{L^2}^p|/e_{L^2}^p$, where $e_{L^2}^m = \|u - u_h^m\|_{L^2}$ is the L^2 -norm of the mapped error, and $e_{L^2}^p = \|u - u_h^p\|_{L^2}$ is the L^2 -norm of the physical error.

	H^1 -error mapped vs physical			
Refinement level	Lagrange $k = 1$	Lagrange $k = 2$	Hermite $k = 3$	Hermite $k = 4$
0	1.324×10^{-16}	1.478×10^{-15}	4.964×10^{-5}	1.459×10^{-7}
1	2.571×10^{-16}	1.211×10^{-14}	2.914×10^{-5}	1.866×10^{-6}
2	1.276×10^{-16}	1.151×10^{-13}	8.914×10^{-6}	2.416×10^{-6}
3	3.820×10^{-16}	2.369×10^{-12}	2.881×10^{-5}	2.607×10^{-6}
4	3.691×10^{-15}	1.303×10^{-11}	3.671×10^{-5}	2.773×10^{-6}

Table 5: $|e_{H^1}^m - e_{H^1}^p|/e_{H^1}^p$, where $e_{H^1}^m = |u - u_h^m|_{H^1}$ is the H^1 -seminorm of the mapped error, and $e_{H^1}^p = |u - u_h^p|_{H^1}$ is the H^1 -seminorm of the physical error.

9.3 Note: Extended space approach

1. Choose dof set Λ , constraint set $\mathcal{C}_0$. Choose ℓ so that we have unisolvency of the dofs over $\mathbb{P}_\ell(E)$. Pick basis $\mathcal{B}_0$ for $\mathbb{P}_\ell(E)$.
2. Extend dof set to Λ^* , to enforce L^2 -projection:

$$\Lambda^* = \Lambda \cup \left\{ v \mapsto \frac{1}{|E|} \int_E vm : m \in \mathbb{P}_\ell(E) \setminus P_{\mathcal{C}_0} \right\}, \quad (9.2)$$

where $P_{\mathcal{C}_0}$ is the space already prescribed by the constraints, perhaps $\mathbb{P}_{\ell-2}(E)$.

3. Define extended space $\tilde{V}^*$ in the normal way, including edge conditions and problem conditions, ensuring it contains $\mathbb{P}_\ell(E)$.

For example,

$$\tilde{V}^* = \{v \in H^1(E) : \Delta v|_E \in \mathbb{P}_\ell(E), v|_{\partial E} \in \mathbb{B}_\ell(\partial E)\}. \quad (9.3)$$

4. Compute value projection Π_0^E on $\tilde{V}^*$ from $(E, \Lambda^*, \mathcal{C}_0, \mathcal{B}_0)$. Check if it is admissible.
5. Define

$$\tilde{V} := \left\{ v \in \tilde{V}^* : \lambda(v) = \lambda(\Pi_*^E[v]), \lambda \in \Lambda^* \setminus \Lambda \right\}. \quad (9.4)$$

6. Use Π_0^E and similar for edge projection on the smaller space. Now it is the full L^2 -projection up to order ℓ , instead of just on the constraint space.

Lemma 9.2. $\mathbb{P}_\ell(E) \subset \tilde{V}$.

Proof. Π_*^E admissible so $\Pi_*^E[m] = m$ for all $m \in \mathbb{P}_\ell$.

So $\lambda(m) = \lambda(\Pi_*^E[m])$ for all $\lambda \in \Lambda^* \setminus \Lambda$.

So $m \in \tilde{V}$. □